

# **INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION**

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## **APPROVAL SHEET**

This undergraduate thesis entitled INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION, prepared and submitted by Ashanty Madeline DC. Cruz, Brian Ezekiel D. Batalon, Mark Jayson F. Villacura, and Tristan Manuel T. Sto. Tomas, in partial fulfillment of the requirements for the degree Bachelor of Science in Electronics Engineering, is hereby recommended for approval.

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**-The Researchers**

## DEDICATION

*This study is wholeheartedly dedicated to our beloved parents, who have been our source of inspiration, who believed in us, and gave us strength when we thought of giving up, and who continually provide their moral, spiritual, emotional and financial support. To our siblings, relatives, mentors, and friends who shared their words of advice and who have witnessed all of our hardships and provided encouragement to finish this study. And lastly, we dedicate this piece of hard work to the Almighty God, to whom we have gotten our strength, wisdom, understanding, and inspiration in accomplishing this work.*

*-Ashanty, Brian,  
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## **ABSTRACT**

|                               |   |                                                                                                                   |
|-------------------------------|---|-------------------------------------------------------------------------------------------------------------------|
| TITLE                         | : | IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection                        |
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### **EXECUTIVE SUMMARY:**

The general objective of this study entitled IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection is to optimize poultry incubation by integrating automation, computer vision, and renewable energy. The research involves variables such as chicken egg fertility detection accuracy, temperature and humidity regulation, and remote monitoring through a web-based application. This study was conducted within the academic year 2024-2025 at the University of Rizal System.

The research sought to answer key questions related to optimizing the fertility detection and incubation processes. This study explores the accuracy of the system's temperature and humidity regulation mechanisms and evaluates the

effectiveness of the YOLO deep learning model in detecting fertile and infertile chicken eggs. Additionally, it assesses the overall acceptability of the system among end-users and experts based on ISO/IEC 25010 standards. Statistical treatment included weighted mean for survey data.

The research employed a developmental and descriptive research design to analyze and enhance the functionality of the IncuVision system. A deep learning model was trained on historical dataset of fertile and infertile eggs, with 3,180 annotated images. Additionally, temperature and humidity data were gathered from sensors, and system acceptability was evaluated through a Likert-scale survey conducted among 32 respondents, consisting of 7 experts and 25 potential users. The system is developed using a Raspberry Pi 4B as the main control board, integrating a temperature and humidity sensor, an automatic egg-tilting mechanism, and a computer vision-based fertility detection system using the YOLOv8 deep learning model, with a web-based interface for remote monitoring and control. Research instruments include accuracy tests for sensor for incubation conditions, percent error analysis for performance evaluation of fertility detection model, and a Likert-scale survey to assess system acceptability.

The system's sensor for measuring the temperature and humidity demonstrated accurate readings, ensuring stable incubation conditions crucial for embryo development. The use of a Raspberry Pi 4B for processing, along with a 100W solar panel and a 12V lead-acid battery for backup power, enhanced system reliability and accessibility. Real-time monitoring and notification features provided users with a seamless incubation management experience via a user-friendly web-

based application. Testing and evaluation confirmed that IncuVision effectively optimizes incubation conditions and accurately detects egg fertility. The system's fertility detection showed a low percentage error, indicating a high accuracy and surpassing the set threshold, demonstrating its reliability in fertility classification.

The results of the acceptability survey showed that the IncuVision system was rated as very much acceptable in all evaluation criteria, including functional suitability, usability, performance efficiency, reliability, and maintainability. The result affirms that IncuVision meets industry standards and user expectations, making it a viable solution for poultry incubation.

The findings confirm that the developed system effectively optimizes poultry incubation by automating the candling and incubation processes. Through the integration of computer vision and renewable energy, it enhances precision, efficiency, and sustainability. The deep learning model achieved high fertility detection accuracy, minimizing the need for manual candling. The system also maintained stable temperature and humidity levels, ensuring optimal conditions for embryo development and received strong user approval, validating its effectiveness in modern poultry farming.

Based on the findings and conclusion, the following recommendations were drawn up. The researchers recommend improving incubator ventilation and sealing to prevent air leaks, expanding the AI model training dataset to increase detection accuracy, enhancing scalability by increasing egg capacity, upgrading the processor to improve system performance, and using non-reflective or darker egg trays to reduce light reflection during fertility detection.

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## Chapter 1

### **BACKGROUND OF PROJECT**

This chapter covers the background of the study, objectives of the study, theoretical and conceptual framework, the significance of the study, the scope and limitations of the study, and the definition of terms.

#### Background of the Study

Poultry farming is important in agriculture worldwide since it is one of the major sources of protein consumed by humans. Despite the critical role of poultry in the global food supply, the poultry industry faces significant challenges in meeting the growing global demand for eggs and chicken meat. Traditional incubation methods, which rely on manual intervention in handling incubators and conducting the candling process, have proven inefficient and prone to error, leading to lower hatch rates and declining broiler stocks. These inefficiencies not only compromise production but also contribute to higher poultry prices.

Reflecting the global scale of these challenges, over 50 billion chickens are raised for egg and meat production, a figure that continues to rise as the world population is expected to hit 9.6 billion by 2050. The poultry industry has experienced remarkable growth in recent years, with poultry meat production surging by 121 percent and egg demand increasing by 65 percent, significantly outpacing growth in the pork and beef sectors. In 2017, Canada exported 39 million hatching eggs to the US mainly valued at over \$68 million whereas importing more than 141 million hatching eggs for broilers from the US, valued at more than \$49 million. These statistics underscore the need for innovative farming practices,

including smart incubation systems where technology is essential for improving hatching success, ensuring poultry welfare, and highlight how international trade in hatching eggs exemplifies the global interdependence of the poultry industry as it adapts to meet the rising demand for animal-based food (Agriculture and Agri-Food Canada, 2017).

In the Philippines, the Philippine Statistics Authority (PSA) reports that, as of September 2023, the annual growth rate of broiler and layer chickens has been declining. As a result, the supply of broiler and layer chickens decreased by 2.0 percent and 1.0 percent, respectively, as shown in Appendix H. Despite this decline, CALABARZON remains one of the highest egg-producing regions, contributing 40 percent of the country's total egg production.

According to an article published by Veterinaria Digital (2022), chicken accounts for 93 percent of global egg production, making eggs one of the most consumed animal-based products. Reflecting this global significance at the national level, poultry production in the Philippines ranks as the second most important agricultural subsector. Limited supply, coupled with high pork prices driven by the African Swine Fever (ASF) outbreak, has led consumers to shift their preference from pork to chicken, solidifying chicken as the meat of choice for Filipinos.

One of the primary reasons for the decline in poultry production is the prevalence of diseases such as Inclusion Body Hepatitis (IBH), which have posed significant challenges for producers. In response, the Department of Agriculture (2021) issued special import permits for autogenous vaccines to help combat these

diseases. IBH, caused by a resilient fowl adenovirus, has directly contributed to a reduction in laying hens and an increase in egg prices.

The challenges stemming from poultry diseases and shortages in the supply of poultry products have negatively impacted the market. In Metro Manila, the chicken prices reached a record high of 250 pesos per kilogram as local broiler production declined and farmers endeavored to recover losses incurred between January and June 2024. According to the Department of Agriculture's July 2024 price monitoring report, whole chicken prices ranged from 190 to 250 pesos per kilogram 30 pesos higher than in June 2024. At a press briefing, Jose Gerardo Feliciano, president of the United Broiler Raisers Association (UBRA), attributed the rising prices to reduced broiler stocks following farmers' efforts to decrease flock sizes to mitigate earlier losses (Arcalas, 2024).

A key component in improving poultry production is optimizing the incubation of eggs. Madkour (2021) explains that natural incubation, also known as brooding, involves a broody hen providing warmth and care for eggs, which is suitable for hatching a small number of eggs. However, Okpagu and Nwosu (2016) emphasize that effective incubation requires managing temperature, humidity, ventilation, and egg rotation; poor regulation of these factors can negatively affect embryo development. Adame and Ameha (2023) further elaborate that while natural incubation is practical and cost-effective for small-scale operations, it demands time and effort, making it less viable for larger-scale production. They suggest that artificial incubation methods, though requiring higher investment, provide efficiency and consistency by precisely controlling the

incubation environment, leading to better hatchability rates and chick quality.

In relation to this, candling eggs is a crucial aspect in the incubation process, it allows producers to assess fertility and embryonic growth, ensuring that the incubation environment is effectively supporting viable eggs. Vargas et al. (2018) noted that traditional egg candling is a non-destructive internal analysis technique that essentially entails manually passing light through eggs to detect defects. This process makes the eggshell transparent. Also, Dela Cruz and Fadchar (2020) stated that the Philippine poultry sector currently relies on the traditional egg candling procedure which is subjective and laborious. This addresses the needs in automated egg candling system, such as increasing detection speed and accuracy for fertility detection.

Incubating chicken eggs also involves the use of light as a source of heat. Archer (2017) and Li et al. (2021) had a similar finding in their studies, they demonstrate that exposing eggs to light during incubation positively influences embryo development by regulating temperature within the eggs, likely due to increased melatonin secretion, while El-Sabroun and Khalil (2017) found that light enhances chick vitality and hatch weight, although no significant differences in hatchability were observed. Furthermore, light serves as an effective heat source in modern incubators, as evidenced by Rehman (2016), who demonstrated that an 80-watt bulb, can effectively provide heat within incubators.

However, it is crucial to use a camera for image processing in making such automated incubation with egg candling for fertility detection. Zhu (2022) defined image processing as the optimization of camera images using various techniques,

including scaling, color adjustment, and brightness enhancement, to improve presentation and usability. Prijatna et al. (2018) showed that light can have a substantial impact on the accuracy of image processing, with variations in light intensity affecting crucial features like color and area. Any additional light source, like the heating light, can introduce undesirable fluctuations into the collected photos of a chicken egg incubator, so compromising the candling process and precision. More importantly, Hasinoff et al. (2016) and Zhang et al. (2019) noted that too much light might overwhelm camera sensors, producing noise and distorted images, especially in low-light situations where accurate image capture is essential. Thus, in order to guarantee precise fertility detection, the heating light has to be off and the candling process in the dark with only the controlled candling light being used for clear image enhancement.

To keep maintain the heat in incubation process without using of light. Andoh et al. (2022) showed how a fan-powered machine can be used to transfer heat into the incubator room to ensure stability during poor weather or nighttime. In addition, Bansal et al. (2015) stated a fan with heater ensures stable and reliable control because there will be no temperature fluctuations. In contrast, the use of a light bulb as a heat source will lessen the precise temperature control and often provide too bright of lighting which could interfere with candling and image processing. Thus, heaters with fans attached are a more reliable option for keeping the conditions required for incubation best met.

Maintaining heat is important in incubation process and it requires precise temperature for embryo development of chicken egg. In recent studies such as



Yalcin et al. (2022), they emphasize that optimal hatchability and superior chick quality are possible only at temperatures between 37 and 38 degrees. Similarly, De Almeida et al. (2016) further illustrate the variations in incubation temperatures low (36 degrees), control (37.5 degrees), and high (39 degrees) significantly impact skin and feather characteristics vital for temperature regulation later in life, with high-temperature hatchlings exhibiting thinner, more vascularized skin and a preference for warmer environments. Overall, these studies give an indication on the complex interrelation between incubation temperature and chick health in general and its importance in achieving optimal development and performance. Furthermore, Tejeda & Landreth (2022) describe the term physiological zero as a critical parameter that at a temperature below (12-20 degrees), the embryo becomes dormant, and development halts. Only above this temperature will resume the development of the embryo.

Humidity is also one of the important parameters present in incubation process. Maulana et al. (2023) emphasize the importance of 50-60 percent humidity during incubation leading to better hatchability. Poor humidity control can result in variations that affect the integrity of the incubation process. Likewise, Dutta and Anjum (2021) state that improper humidity, along with fluctuations in temperature, negatively impacts hatching success. Furthermore, Oliveira et al. (2020) state that it is very important to keep the humidity at the right level because if the humidity goes up and down too much then the hatchability and the whole incubation process will be affected.

On the other hand, based on Bilalissi et al. (2022), ventilation during

incubation plays a huge role in embryonic development and post-hatch performance, especially with pre-incubation storage. In their experiment, they found that eggs hatched in non-ventilated (NV) incubators had a higher hatchability and better post-hatch chick weight than those hatched in ventilated (V) incubators. Non-ventilation, particularly in the case of eggs held for extended periods of time, seemed to offset the negative influences of storage on embryonic development, which reflects the overall positive influence of reduced ventilation on hatchability. Similarly, Fares et al. (2023) showed that unventilated periods in the first part of incubation enhanced hatchability and chick size due to the accumulation of CO<sub>2</sub> inside the incubator. They found that a gradual increase in CO<sub>2</sub> in the first few days of incubation increases hatchability and reduces hatch window, again proving how critical ventilation times and egg positioning are in incubation. Furthermore, Okur and Eratalar (2021) determined that eggshell thickness, not the ventilation program, had a greater influence over hatch time and that those eggs of medium thickness hatched the best.

The researchers also consider the effects of storage duration on the hatchability rate and mortality rate of the chicken egg inside the incubator. According to Archer and Cartwright (2017), it is crucial to incubate eggs as soon after collection as possible, ideally within 7 to 10 days of being laid. They found that hatchability declines rapidly when eggs are stored beyond this timeframe, decreasing by 0.5 to 1.5 percent for each additional day in storage. This observation is supported by Addo et al. (2018), who noted that extended storage durations adversely affect egg quality, leading to higher embryonic mortality and

reduced hatchability. Similarly, Melo et al. (2020) emphasized the negative effects of prolonged storage, indicating that eggs stored for 12 days experienced increased late embryonic mortality, regardless of whether they were turned during storage. Moreover, research by Abioja et al. (2020) demonstrated the key quality parameters, such as yolk diameter and albumen weight, significantly declined with longer storage durations. Their findings revealed that storage beyond eight days resulted in increased embryo mortality and decreased hatchability rates.

The tilting of eggs during incubation is also a factor that affects both hatchability and the overall health of chicks. Aigboje et al. (2023) introduced a low-cost electric egg incubator designed to optimize incubation conditions, featuring an integrated turning mechanism. Their findings revealed an incubation temperature range of 36.2 to 38 degrees celsius, resulting in fertility and hatchability rates of 81.7 and 58.3 percent, respectively. Similarly, Sunday et al. (2020) constructed an automated incubator utilizing a tilting tray mechanism that tilted eggs at a 40° angle every hour for four minutes, achieving a hatchability rate of 89.47 percent. The management of tilting angles and intervals is essential for enhancing embryo development by promoting gas exchange and preventing adhesion of the embryo to the eggshell, as indicated by these studies. Furthermore, Amdadul et al. (2021) stated that egg turning should occur regularly throughout incubation because it supports proper embryo development. Higher hatchability rates result from the embryo prevention from sticking to the shell membrane along with balanced nutrient distribution and optimal gas exchange.

With the parameters and features such as temperature, humidity,

ventilation, and tilting of eggs, recent studies have introduced various innovations in egg incubation such as Sobejana and Bacalso (2021) where they developed a poultry egg incubator with integrated temperature and humidity controls and SMS notifications, addressing issues of manual incubation and improving overall efficiency. In addition, Berry (2017) detailed essential aspects of artificial incubation, including temperature, humidity control, and ventilation, all of which are crucial for successful egg development.

Moreover, advancements in incubation technology continue to drive improvements in poultry production efficiency, addressing key challenges in the industry. Bacalan et al. (2019) emphasize that on-site business expertise and situational experience play a crucial role in the success of technology incubators, particularly in developing countries. These factors contribute to better resource allocation, improved policy formulation, and the effective implementation of incubation systems. In line with this, Maroma et al. (2023) introduce an automated incubator equipped with a backup power supply, which effectively mitigates the risks associated with power outages. This technological advancement not only enhances hatch rates but also ensures a more stable incubation process, reducing losses in poultry farms that frequently experience electricity disruptions.

Furthermore, Tadeo (2024) explores the impact of genetic and environmental factors on the performance of Bolinao native chickens, providing valuable insights into optimizing breeding strategies. Understanding these factors allows farmers to make informed decisions regarding selective breeding, ensuring that the best traits are passed down to future generations. Complementing these

findings, Boleli et al. (2016) review the historical evolution of incubation practices, underscoring the importance of optimizing critical factors such as temperature, humidity, and ventilation to maximize hatchability and improve chick quality. These studies collectively highlight the ongoing improvements in incubation technology, demonstrating how scientific advancements and practical innovations contribute to increased efficiency, sustainability, and productivity in the poultry industry.

Building on this, technological advancements have significantly improved the incubation process, enabling better precision and control over hatching conditions. Niranjan et al. (2021) developed a smartphone platform-based incubator that uses an Internet of Things (IoT) interface to monitor temperature, humidity, and egg turning, enhancing efficiency through real-time data tracking and remote accessibility. However, its effectiveness is influenced by environmental stability and connectivity limitations, which can impact performance. Despite these challenges, the study underscores the potential of IoT-driven solutions in modern incubation, paving the way for further research to optimize automation and improve hatch rates.

Furthermore, other remarkable innovations in detecting egg fertility through image processing and artificial intelligence include automation to increase effectiveness and accuracy, such innovations using computer vision with You Only Look Once (YOLO) model. According to Voulodimos et al. (2018), computer vision is a field that leverages deep learning techniques that allow systems to perform tasks like object detection, face recognition, and activity monitoring, enhancing their ability to understand and interact with the visual world effectively. While, Jiang

et al. (2022) describe YOLO as a fast and efficient object detection algorithm that processes images in real-time, distinguishing it from traditional methods like Convolutional Neural Networks (CNNs). Its ongoing improvements aim to optimize target recognition and feature extraction for diverse applications.

To exemplify, Febriani (2024) applied the model called YOLOv9-S for real-time detection and classification of chicks. It focused on identifying hatched eggs and monitoring chicks' activities rather than detecting egg fertility, but it still provided vital insights into applying deep learning in incubation. This highlights the potential of YOLO models in automated systems and their ability to improve efficiency and accuracy in monitoring. This proves that YOLO can enhance automation and decision-making processes in incubation with minimal manual intervention and efficient environmental control to improve hatch rates.

On the other hand, Potrimba (2023) explains Region-based Convolutional Neural Network (R-CNN) as a type of deep learning architecture that focuses on detecting objects in specific regions of interest in an image. For instance, Tolentino et al. (2022) developed an automatic incubator using a Region-based Convolutional Neural Network (R-CNN) to detect the maturity of chicken and duck eggs with a classification accuracy of 80.5 percent. Despite having significant accuracy, the challenges faced by most computer vision technologies in detecting egg fertility is due to low image quality and scattered candle lights, which affected the view of the egg's embryo. Another factor is that R-CNN draws a rectangular shape that outlines the egg which overlaps with other neighboring eggs, thus reducing accuracy.

Hence, the researchers came up with a study that combines incubator and computer vision entitled IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection which is used to monitor and predict the status of the chicken egg fertilization process leading to the successful hatching and production of chicken. In addition to this, the researchers considers the following rules and regulations or the requirements of Philippine National Standards. The Philippine National Standard along with the Bureau of Agriculture and Fisheries Standards set an Egg Incubator Specifications to meet the required and effective hatching process of poultry eggs. Under Section 7 Performance Requirements of Egg Incubator Specification, the table shown in Appendix B includes that the Philippine National Standard Egg Incubator Specification shall attain the relative and temperature criteria for poultry eggs.

The Philippine National Standard also stated in Egg Incubator Specification, under section 6: Fabrication Requirements, 6.4: Turning System, 6.4.2 & 6.4.3:

“For automated egg incubators, eggs shall be rotated or tilted according to the manufacturer’s specifications.  
For the tilt-type egg incubator, the eggs shall be tilted at an angle of 35° to 45°.”

This section ensures proper embryo development and prevents the embryo from sticking to the shell leading to deformities and dropping the hatch rates. Also, ensures even heat distribution, and improves oxygen exchange.

For the Ventilation System, the Philippine National Standard Egg Incubator Specification, under the statement of section 6.3: Ventilation System;

“The egg incubator should contain an air-circulating fan and exhaust mechanism (e.g., fan or ventilation holes) to allow airflow inside the incubator and maintain a uniform temperature and relative humidity for incubation.”

This is to provide a proper process of incubating chicken eggs and meet even the minimum requirements of the Philippine National Standard on making such advanced chicken egg incubators.

In summary, the poultry industry meets rising world demand for chicken meat and eggs mainly resulting from population growth and supply deficits, most prevalent within the Philippines. Traditional incubation methods have inefficiencies related to limitations in detection regarding the fertility of the eggs, thus compromising hatch rates and severely eroding the number of broiler stocks on which prices rise eventually. Such challenges identify a need for high priority for new technologies designed to increase efficiency in chicken production. IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection significantly fill the existing gap by simply bringing superior technology to support the optimization of fertility detection and incubation processes, thereby allowing the poultry industry to better meet upward demand.

#### Objectives of the Study

This study aims to develop an enhanced technology of chicken egg incubator with computer vision to optimize the fertility detection and incubation processes of chicken eggs.

Specifically, this study sought to:

1. Design a chicken egg incubator with computer vision for fertility detection with the following features:

- 1.1 Web-based Monitoring:

- 1.1.1 User-Friendly Interface; and



- 1.1.2 Notification Alert System;
- 1.2 Automatic Regulation with the following parameters:
  - 1.2.1 Temperature;
  - 1.2.2 Humidity; and
  - 1.3 Solar Power for Backup Power Source.
- 2. Test the accuracy of sensor readings of the chicken egg incubation system in terms of:
  - 2.1 Temperature; and
  - 2.2 Humidity.
- 3. Determine the Percent Error of the Automated Chicken Egg Fertility Detection System with Reference to Traditional Candling.
- 4. Determine the level of acceptability of the developed incubator with reference to ISO/IEC 25010:
  - 4.1 Functional Suitability;
  - 4.2 Performance Efficiency;
  - 4.3 Usability;
  - 4.4 Reliability; and
  - 4.5 Maintainability.

### Theoretical Framework

The theoretical foundation of this study is anchored on the Principles of Deep Learning Theory, Theory of Light, and Psychrometry. This offers an understanding of technology and environmental science in the context of chicken egg incubator.

Roberts and Yaida (2021), the author of the book entitled *The Principles of Deep Learning Theory*, explains that deep learning is a field of artificial intelligence that uses artificial neural networks to mimic the functions of the human brain. These artificial neural networks are designed as flexible systems made up of many small units called neurons, which allows the deep learning model to be trained, learn, and solve real-world problems. Deep learning comes in many models that can handle complex data making them useful in tasks involving visual data, such as image recognition, object detection, and classification. Deep learning models have a wide range of applications such as computer vision, natural language processing, and speech recognition.

The principles of deep learning are essential for developing models that uses computer vision to classify fertile and infertile chicken eggs. By leveraging advanced algorithms, this approach can enhance accuracy and speed in the egg classification process.

Yuan (2024) discusses the Theory of Light, explaining that light is an electromagnetic wave that undergoes reflection and refraction upon encountering different media. Some light is reflected back into the original medium, while the rest refracts and changes direction as it passes through various materials. These principles are crucial in analyzing the internal structures of objects, such as chicken eggs. When an LED light illuminates an egg, the scattering, absorption, and transmission of light depend on the egg's internal composition. Fertile eggs, containing developing embryos, absorb and scatter more light, appearing darker, whereas infertile eggs allow more light to pass through, appearing clearer.

The Theory of Light is crucial in this study as fertility detection relies on optical properties to differentiate fertile from infertile eggs. The interaction between light and the egg's internal structure enhances visibility and classification accuracy, enabling a non-invasive and reliable method for distinguishing egg fertility.

With regard to this, there is also the need to assess the environmental conditions for chicken eggs and in this respect, psychrometry offers the right platform for determination. Psychrometry is the study of the physical and thermodynamic properties of moist air such as humidity, temperature, and moisture content. Fabian (2023) notes that psychrometry is essential for efficiently functioning heating, ventilation, and air conditioning or HVAC systems. Components of psychrometry include dry bulb temperature, wet bulb temperature, relative humidity, and dew point temperature. These elements aid in studying the effect of moisture in the air on the temperature and comfort, therefore making it important in managing the environmental conditions within any setting.

This is vital for creating an ideal environment for egg development. The right balance of temperature and humidity influences the incubation process, affecting hatchability and chick health. By applying psychrometric data, incubator designs can be optimized to ensure suitable conditions for hatching eggs.

### Conceptual Framework

The conceptual framework of this study, as shown in Figure 1, is based on the Agile System Development Life Cycle (SDLC) Model. This approach ensures an iterative and flexible process, allowing continuous improvements through feedback and testing.



**Figure 1**

Conceptual Framework of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

Firstly, the data gathering is conducted, collecting key information from various sources, including related studies on incubation and fertility detection, journals, articles, datasets, and online resources that help in the design and development of the study. These sources provide insights into the latest developments in computer vision and automation technologies. Interviews with experts in poultry farming and incubation, as well as consultations with potential users, also play a significant role in understanding the system's real-world applications and identifying user needs. During the planning stage, the research objectives are clearly defined, focusing on the integration of computer vision for fertility detection in an enhanced chicken egg incubator. In the designing phase, a detailed conceptualization of the system is carried out. This includes creating schematic diagrams of the electronic components, developing flowcharts that outline the operational processes of the egg incubator, and creating a prototype design. The design incorporates the use of a deep learning model to enable fertility detection.

The development phase follows, where the physical assembly and construction of the incubator is carried out, along with the programming required for maintaining incubation conditions and for fertility detection based on deep learning algorithms. This stage also involves training the deep learning model using datasets of chicken eggs to ensure accurate fertility detection. Finally, in the testing phase, the trained deep learning model is tested for accuracy in detecting egg fertility. The entire system, including the incubator's features and fertility detection capabilities, undergoes thorough testing, debugging, refining, and

performance evaluation. Moreover, a questionnaire checklist is also prepared, validated, and administered to gather feedback from end-users and experts.

The output is the developed IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection. Lastly, the feedback is used for further system improvements until the product can be launched and implemented. This system represents a significant innovation in poultry farming, providing a reliable and automated method for detecting egg fertility, which enhances the efficiency of the incubation process.

### Significance of the Study

This study aims to design and develop an Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection, which will help improve poultry farming practices, reduce manual labor, accurately monitor the egg conditions, and improve hatchability rates. Accordingly, this study intends to benefit the following:

Poultry Farmers. The enhanced incubator system will assist farmers by automating the egg candling and turning processes, improving the accuracy of fertility detection, and reducing manual labor, leading to increased productivity and healthier hatchlings.

Poultry Products Consumers. As the main consumers of poultry farm products, the community will benefit indirectly through a more stable supply of these products. The stability will enhance food security and support local economies by ensuring consistent availability and quality of poultry goods.

Future Researchers. The knowledge gained from this study will help future researchers further innovate the technologies related to poultry farming. It will also

serve as a milestone in their forthcoming studies.

### Scope and Limitations

This study, IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Predictive Fertility Analysis, was conducted by fourth-year students of the Bachelor of Science in Electronics Engineering at the University of Rizal System – Morong Campus during the academic year 2024-2025. The rationale behind this study is to address the inefficiencies of traditional incubation methods and improve fertility detection, thereby enhancing hatch rates and overall poultry productivity.

The study involves the development and evaluation of the prototype of an automated incubator that integrates computer vision and deep learning for real-time fertility detection. The incubator is designed to monitor and regulate temperature and humidity automatically, classify eggs as fertile or infertile using a YOLO deep learning model, provide notifications for infertile eggs, support solar power as a backup power source, and enable real-time monitoring through a web-based application.

The scope of this study also includes the 21 days incubation of a total of twenty-four (24) Rhode Island Red chicken eggs sourced from Buenvenida Farm in Boso-Boso, Antipolo City. Additionally, the researchers collected 200 images during the candling day of the incubation process, including those captured at Honey Red Candice Hatchery in Morong, Rizal, for the training and testing of the deep learning model. Along with these, 2,980 images from Roboflow were used, resulting in a dataset of 3,180 images. The YOLO model is trained using this

dataset, and its built-in accuracy results are analyzed to evaluate its effectiveness.

Despite its innovations, the study has limitations. It is restricted to a small sample size, which may limit generalizability to other breeds or large-scale incubation. As a prototype, it focuses on system functionality rather than mass production or commercial scalability. The accuracy of fertility detection is influenced by image quality, lighting conditions, and dataset limitations, indicating that further training with a more diverse dataset could improve performance. The study employs fixed hardware and software configurations, including the Raspberry Pi and YOLO deep learning model, without exploring alternatives.

Additionally, the device can automatically candle and detect egg fertility for a maximum of 24 chicken eggs at a time, which may limit its application for larger-scale poultry operations. The research is also conducted within two academic semesters, restricting long-term performance evaluation across multiple incubation cycles. This study does not cover disease detection, nutritional impacts on hatch rates, or economic feasibility for large-scale poultry farms. These limitations provide opportunities for future research to refine the system, enhance accuracy, and expand its practical application in poultry farming.



### Definition of Terms

The important terms used in this study have been defined operationally for clarification.

Candling. It is a technique used to inspect the fertility and growth of the chicken eggs by shining a bright light into them, thus providing a view of the inside part of the egg.

Computer Vision. It is a sub-branch of artificial intelligence and computer science in which the computer is designed to understand and interpret real-world visual information from images and videos. Computer vision keeps track of the eggs and carry out analyses for factors that may relate to development and health.

Confusion Matrix. It is used for evaluating the performance of the deep learning model for detection of fertility of chicken eggs. Performance metrics of an algorithm are accuracy, precision, recall, and F1 score, which are calculated on the basis of the true positive, true negative, false positive, and false negatives.

Data Collection. It is collection data of fertile and infertile eggs to give notification alert and monitoring to the user.

Fertility Detection. It examines the possibility that the chicken egg is fertilized and has the potential to hatch into a chick. It involves analysis of the egg patterns such as embryo and is often done using complex algorithms.

Functional Suitability. This refers to the ability of incubator to accurately detect and classify chicken egg fertility using YOLLO deep learning model, ensuring it meets the specific requirements for effective monitoring.

Humidity. It refers to the amount of moisture content that is in the air at any given

temperature. In an incubator system, humidity is maintained at a particular desired level to support the proper growth and hatching of the chicken eggs.

Incubator. It is a device designed to maintain optimum environmental conditions, including temperature, humidity, and ventilation, to artificially incubate eggs and facilitate their development until hatching. In this study, the incubator is specifically for chicken eggs, enhanced with advanced monitoring and control features for improved incubation efficiency.

Maintainability. This refers to the ease with which the computer vision system can be updated for serviced to enhance its fertility detection capabilities, ensuring long-term performance.

Notification Alert System. It is a feature of the incubator system that provides alarm and notifications service. This system ensures that the user is always informed about critical events within the incubator, including whether the eggs in the batch are infertile or ready to hatch. Additionally, it alerts users to any abnormal conditions affecting the incubator, enabling timely interventions.

Performance Efficiency. This term describes the effectiveness and speed of the YOLO model in processing images to identify fertile and infertile eggs, optimizing computational resources during detection.

Reliability. Reliability indicates the consistency of the fertility detection system in producing accurate result over time, minimizing false positives and negatives in egg classification.

Solar Power for Backup Power Source. It is a feature of the chicken eggs that keeps the incubator during power outage by the use of solar energy systems, like

solar panels and batteries, to provide electricity and keeps the chicken eggs from dying.

Temperature. This refers to the heat inside the incubator which supports the development of the egg. The control system ensures the temperature is maintained at the appropriate level to facilitate proper growth and hatching.

Usability. Usability emphasizes how intuitive and accessible the web-based interface is for user to monitor fertility detection results, ensuring they can easily interpret the data.

User-Friendly Interface. The incubator control and monitoring system is intuitively designed in a user-friendly manner. This interface allows users to interact with the incubator settings effortlessly, eliminating the need for advanced technical skills.

Web-based Monitoring. It is a feature of the incubator system that provides remote access for users through the web interface, whereby the status of the incubation process is observable. It allows immediate monitoring and management of the incubator from any device connected to the internet.

YOLO or You Only Look Once. It is an algorithm used for detection, it works by analyzing the entire image in one go, predicting chicken eggs and classifying whether it is fertile or infertile all at once, rather than scanning the image multiple times. This makes it fast and efficient, allowing for real-time detection of multiple eggs in a single frame.

## Chapter 2

### **DESIGN METHODOLOGY**

This chapter discusses the research technical design, the research instruments, the procedures and methods of the study, the project design models, and the statistical treatment that is used for this study.

#### Research Technical Design

This study applied the developmental and descriptive method of research.

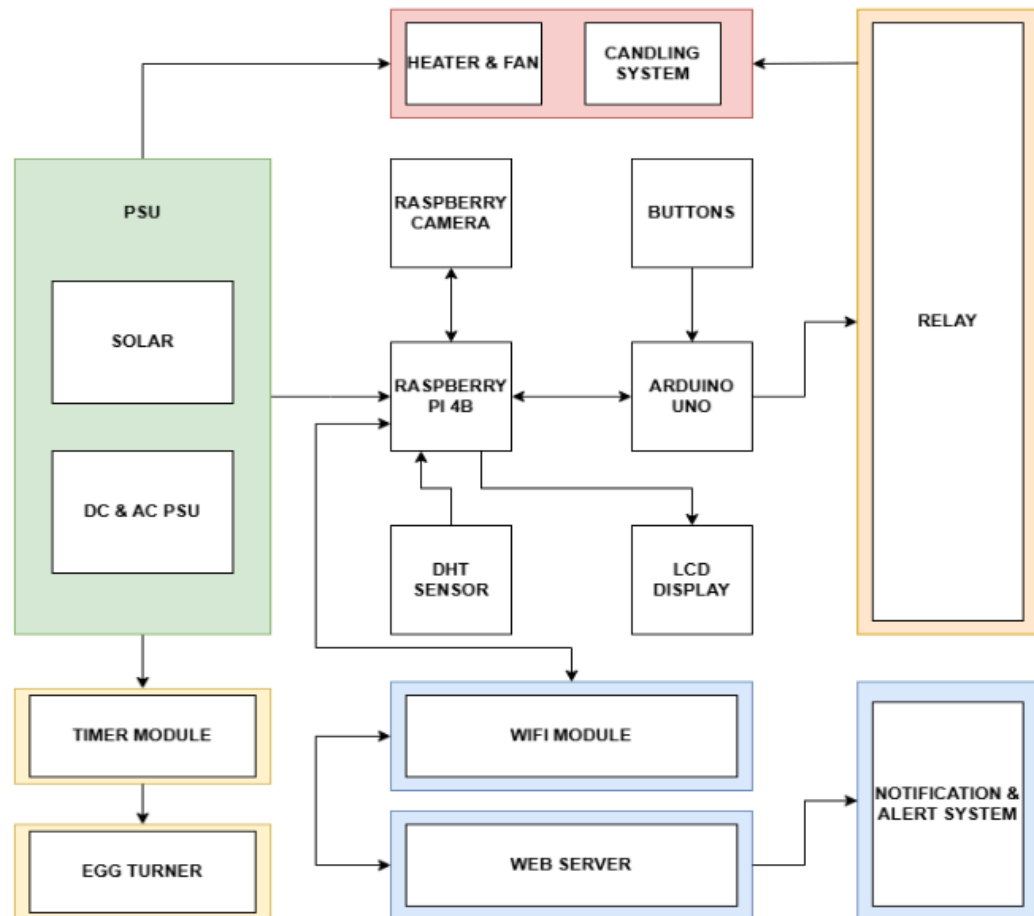
Ibrahim (2016) defined the developmental research as an interactive, cyclic process of development and research in which theoretical ideas of the designer feed the development of products that are tested in classroom tastings, eventually leading to theoretically and empirically founded products, learning process of the developer and (local) instruction theory.

Pimentel (2018) describe descriptive research as a research method used to described the existing phenomena as accurately as possible. The researchers utilized this research design to gather data from the respondents. Furthermore, a questionnaire checklist was utilized to evaluate the level of acceptability of the device in terms of functional suitability, performance efficiency, usability, reliability, and maintainability.

The technical design of the proposed capstone project is illustrated through three key figures. These figures include the block diagrams and flowchart of the system.

Heading to the concept of Figure 2, it illustrates the block diagram of the incubation system, where the Raspberry Pi 4B serves as the main control board.

Connected to it are several key components, including a Digital Temperature and Humidity (DHT) sensor, which monitors and regulates the internal environment of the incubator. The system also features a heater and fan to maintain optimal conditions for egg incubation.



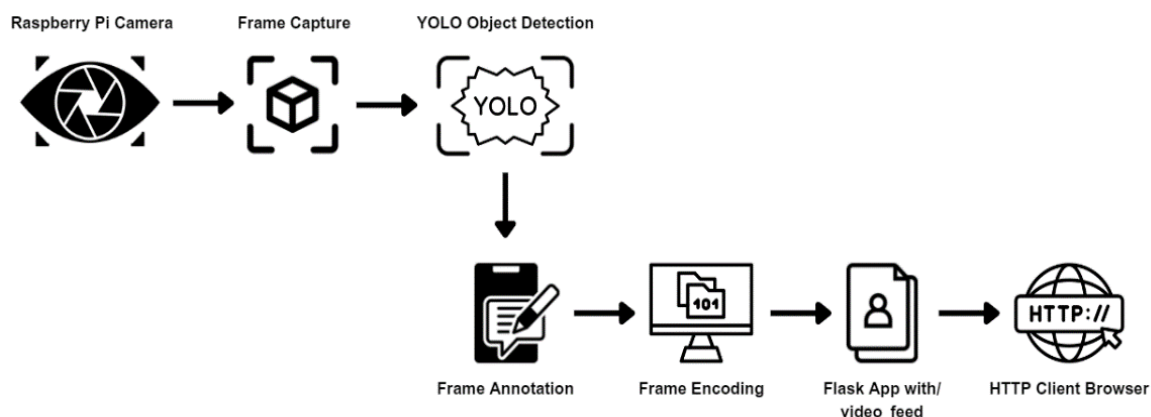
**Figure 2**

Block Diagram of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

Additionally, a Timer Module was set up together with the egg roller mechanism which is integrated to automatically rotate the eggs, ensuring even heating and proper development. An LED candling light is used for egg candling, allowing for the internal inspection of the eggs. Also, a Raspberry Pi camera is

employed to capture images for computer vision analysis, enabling the system to automatically identify whether the eggs are fertile or infertile. The Arduino Uno was used to transfer command to relay which acted as a switch for Heater, Fan, and Candling Lights. Lastly, the Wi-Fi module create connection between the incubator and the user via webserver and it allows the user have access anywhere with built in Notification Alert System via e-mail to help monitor the eggs.

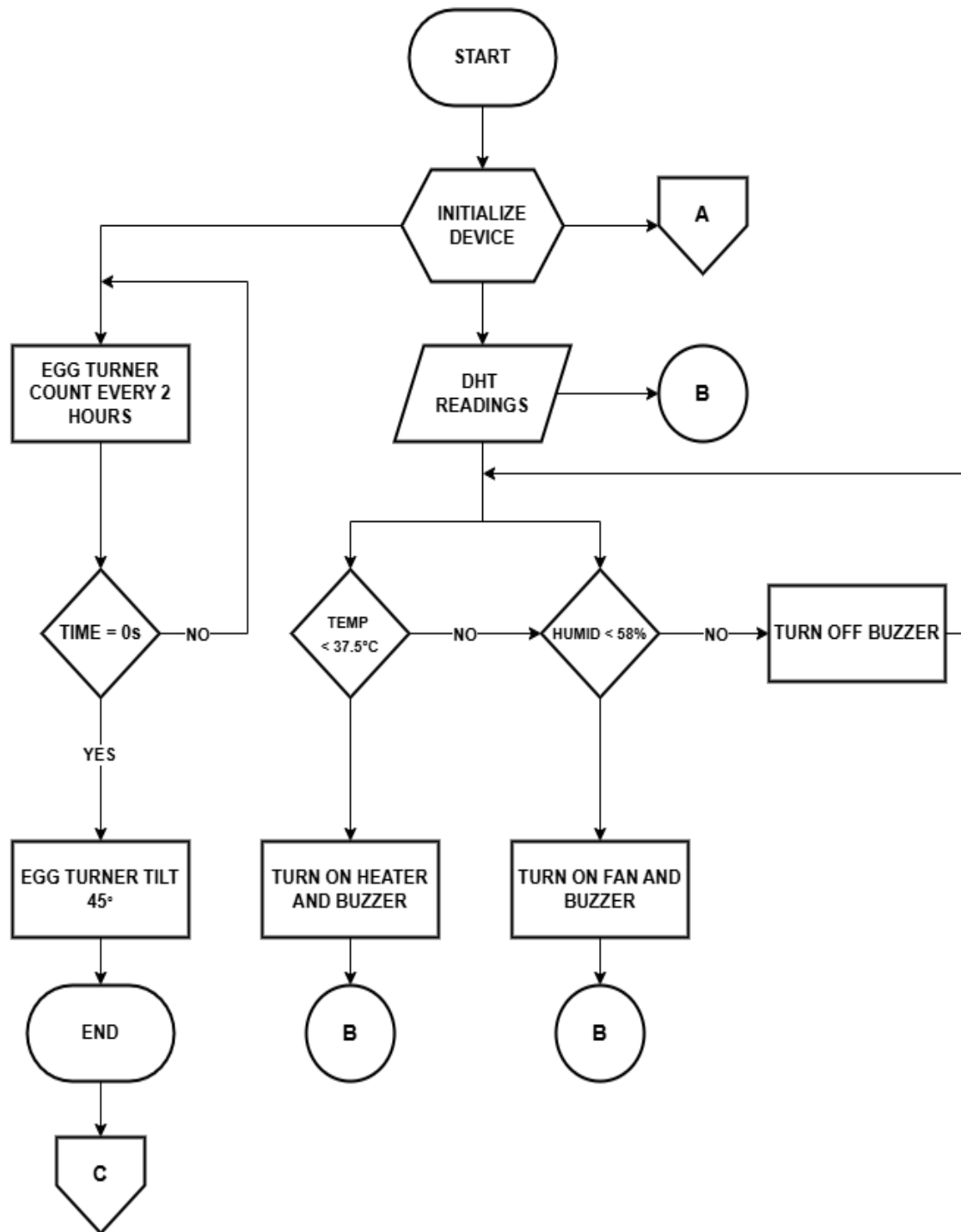
Figure 3 illustrates the computer vision system for egg fertility detection. A Raspberry Pi camera captures live frames, which are processed into images. YOLO detects and classifies eggs based on features like blood vessels or embryo development. Fertility status is then annotated and encoded for efficient transfer.



**Figure 3**

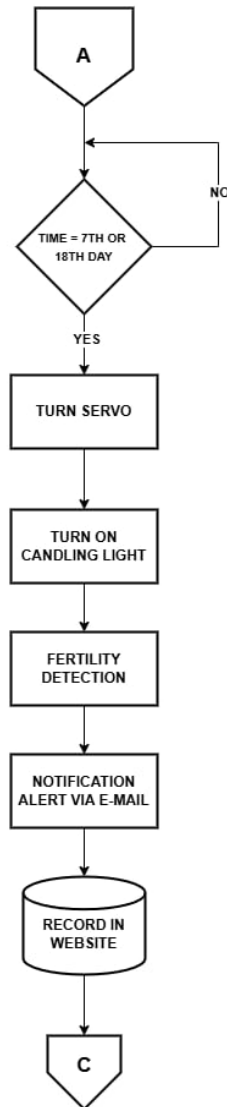
Block Diagram of Fertility Detection Using YOLO Model of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

Moreover, the encoded frames are then streamed through a Flask application that provides a web interface accessible through an explicit route such as video feed. Finally, this processed and annotated video feed was accessed remotely using an HTTP client browser to monitor the eggs in real time.



**Figure 4**

Flowchart for Software of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection



**Figure 5**

Flowchart for Candling System of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

Figure 4 illustrates the flowchart for the incubation system. The process started by initializing the device which includes setting up the user interface. After the boot up, the Raspberry Pi 4B connects to the internet and then proceeds to reads the temperature and humidity via DHT22 sensor. The DHT22 reads if the temperature is less than 37.5 degree Celsius, if yes, then the heater and buzzer



turn on. If the readings on the temperature is not less than the 37.5 degree Celsius, then it turns off the buzzer and proceed to loop back. Humidity readings has the same function as temperature. If the humidity is less than 58 percent, then the relay turns on the fan as well the buzzer. If the humidity is not less than 58 percent, the buzzer turns off and proceed to loop back in DHT readings. The system also has a feature of auto record tracing for the temperature and humidity. For the Egg Turner, the timer module was set to 2 hours and was set by the manufacturers or the researchers, this timer module is not programmable via arduino or raspberry pi therefore a manual method was used, if the time is equal to 0, then the egg tray turns 45 degree for 5 seconds and resetting or ending the process. However, if the time is not equal to 0, the process loop back or the system proceeds to continue the counting.

The flowchart for the software side includes 3 main key-components which also include an off-page reference for candling system as shown in the figure 5. This key-components help stabilize the system throughout the whole incubation process and keeps the user updated anytime and anywhere since the system equipped web-based monitoring for easy access and includes a notification alert via buzzer shown in Figure 4 and via e-mail as shown in Figure 5.

Figure 5 shows the process of candling system. The Raspberry Pi checks if its 7<sup>th</sup> day or 18<sup>th</sup> day of incubating the eggs, if yes, then the servo motor turns, the candling lights turns on, and the Raspberry Pi camera starts detecting fertile and infertile egg. The fertility record was saved in the website and the system notifies the user about the infertile via e-mail.

### Research Instrument

According to Oben (2021), a research instrument is a scientific and systematic designed tool used to collect, measure, and analyze data related to research interests and alignments. Researchers usually need to decide the instrument to use based on the type of study they are conducting: quantitative, qualitative, or mixed-method

This study anchored on the creation of a deep learning model designed to coordinate several system functions including data interpretation, data reporting, and decision making. Historical data on chicken egg fertility was obtained from publicly available databases and supplemented with collected data through the incubator system. This was combined in the training of the model that allowed it to analyze and detect the egg fertility.

Consequently, the researchers had set an accuracy threshold for the deep learning model to ensure it performs effectively and maintains performance comparable to existing computer vision technologies, such as those demonstrated in the study conducted by Tolentino et al. (2022). This depicts that an eighty percent (80%) to one-hundred percent (100%) level of accuracy is described as accurate and acceptable, whereas a lower value or below eighty percent (80%) level of accuracy is interpreted as not accurate and not acceptable.

Moreover, to be able to test the level of accuracy of the DHT sensor in the incubator, the researchers added an analog thermohygrometer and mercury thermometer as reference instruments. A record sheet was used to document real-time temperature and humidity readings from all measurement instruments during

the testing period. Furthermore, for the interpretation of accuracy level, a five percent (5%) margin of error is adhered to according to the specification sheets.

Hence, the researchers utilized a table of the percentage of sensor accuracy and its corresponding interpretation for the study. This depicts that ninety-five percent (95%) to a hundred percent (100%) level of precision is described as accurate whereas a lower value or below ninety-five percent (95%) is interpreted as not accurate.

Additionally, the researchers prepared a general survey checklist containing questions that provides information as per the objectives of the study. The checklist was issued to a random sample of respondents that are poultry experts, and stakeholders involved in poultry farming.

To assess the responses, the study utilized a Likert scale to gauge the level of acceptability of the system. The Likert scale was structured as follows:

| <b>Scale</b> | <b>Range</b> | <b>Verbal Interpretation</b> |
|--------------|--------------|------------------------------|
| 5            | 4.20-5.00    | Very Much Acceptable         |
| 4            | 3.40-4.19    | Very Acceptable              |
| 3            | 2.60-3.39    | Acceptable                   |
| 2            | 1.80-2.59    | Slightly Acceptable          |
| 1            | 1.00-1.79    | Not Acceptable               |

This scale measures the extent to which the system addresses the needs of the target groups.

### Procedures and Methods of the Study

This section describes the activities carried out in order to accomplish the research. It consists of two parts: Conduct of the Study and Conduct of the Design Development.

## A. Conduct of the Study

The researchers followed a series of steps in conducting the study as presented in the Gantt Chart in Appendix A. The first that has been undertaken is conducting a group meeting and identifying the three best research topics. The researchers submitted three titles with their respective concept papers which were subjected to evaluation during the title defense. After evaluation by the research instructor and panelists, the researchers were given the chance to select their preferred topic among the approved titles by the panel. The topic of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection was selected.

After the title defense, the researchers started writing Chapters 1 and 2. In the preparation for the proposal defense, routing, consulting, and signing of the certificate of application to the adviser and panel members are attached to the manuscript of Chapters 1 and 2. Signed documents are then presented for the scheduling of the proposal defense.

In accordance with the conduct of the proposal defense, comments and suggestions by the panel members were noted for the revision of Chapter 1 and Chapter 2. Furthermore, group meetings were conducted usually to take action what are the materials needed for the research project. We also consult and ask people who are knowledgeable about the topic of interest to provide some helpful suggestions and recommendations.

The next actions taken were the prototype development, testing, and evaluation. Data are collected, and results were analyzed for the formulation of

Chapter 3 which includes project design and computations, design results, and analysis and interpretation of the result, and Chapter 4 which consists of a summary of design results, conclusion, and recommendations.

The researchers consulted with their research adviser for the finalization and approval of the manuscript it was submitted for the final defense. The researchers defended their thesis, showcasing the innovation in chicken egg incubation and fertility detection through the developed device. After addressing all the concerns of the adviser and panel members through manuscript revision, the settlement for the hardbound publication was organized and then the copies of the final manuscript were submitted.

#### B. Conduct of the Design Development

The researchers conducted this study entitled IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection with the aim to develop an incubation technology and to improve the process of egg fertility detection using modern tools such as computer vision and deep learning technology. This research provides poultry farmers with a modernized and accessible tool for poultry farming practices.

Firstly, the researchers conducted a comprehensive data-gathering phase. In the planning stage, the researchers determined the technical and functional requirements of the system. The researchers created block diagrams and schematic diagrams, as well as system flowcharts to demonstrate the system operations and interaction of components, outlining a clear set of goals for both the hardware and software components.

The next step involved the careful selection of appropriate hardware components. The researchers chose a high-resolution Raspberry Pi camera for image capturing and a Raspberry Pi 4B (4GB) for the main board to handle the processing needs. Additional sensors for temperature and humidity regulation were incorporated to ensure optimal incubation conditions. On the software side, the researchers utilized Python programming language for deep learning model training and uses OpenCV, TensorFlow, and Pytorch for YOLOV8 model. Labellmg was used for annotating the dataset and Microsoft Visual Studio Code, Javascript, HTML5, CSS, Pi Tunnel is used for hosting server and web application This combination of hardware and software was designed to enable accurate and precise fertility detection and seamless monitoring of incubation parameters.

The main system developed featured a user interface that visualized real-time data on the incubation environment, such as temperature, humidity, and fertility status. The fertility detection algorithm, powered by a convolutional neural network (CNN), analyzed images captured by the camera to identify fertile and infertile eggs with high accuracy. This centralized approach ensured that both environmental monitoring and fertility analysis were managed efficiently, contributing to improved productivity and better outcomes for poultry farmers.

During the whole development process, different stages of testing were carried out to ensure the reliability and functionality of the system developed. This aspect of testing was characterized by continuous debugging and troubleshooting, specifically with the accuracy and precision of the deep learning model and the environmental monitoring and control features of the incubator. The researchers

implemented the system in varying settings while improving the algorithm and the hardware to improve its performance. Errors and inconsistencies in the program of the initial prototype were discovered, which resulted in several iterations of recoding and hardware adjustments to meet the desired performance standards. The level of acceptability of the device was evaluated through feedback from target users, as well as experts in electronics and poultry farming. This feedback played an important role in identifying areas for improvement and further enhancing the system's design. Consequent to the feedback, adjustments were made so that IncuVision met the practical needs of its users. Regular monitoring, maintenance, and data analysis were implemented to support continual improvements, making the system a valuable tool for precision incubation in poultry farming.

#### Project Design Model

In designing the project, relevant scientific and mathematical models were meticulously considered to accurately assess and understand the functioning of each technology involved in achieving the study's objectives.

The study used percent error to assess the performance of the YOLO deep learning model in detecting the fertility of chicken eggs in terms manual and automated.

$$\%ERROR = \frac{|M - A|}{A}$$

Where:

M = Measured Value

A = Reference Value

Measured Value represents the automatic fertility detection of the system using YOLO deep learning model, while Reference Value represents the actual or traditional way of the chicken egg fertility detection.

To determine the temperature of the apparatus, the Steinhart-Hart equation was applied by considering the specific relationship of the NTC thermistor's resistance to temperature characteristics.

$$T = \frac{1}{A + B \ln R + C(\ln R)^3}$$

Where:

T – temperature in degrees Celsius

R – resistance of the thermistor in ohms

A, B, and C – coefficients specific to the thermistor and are provided by the manufacturer

These metrics helped the researchers assess the incubator's performance to guarantee its effectiveness in identifying the chicken eggs' fertility.

#### Statistical Treatment

This study utilized quantitative research technique to evaluate the performance of the developed incubation system.

Weighted mean was computed to analyze survey responses. This provided a better general view of how well the system is meeting the needs of the target groups because the different Likert scale ratings was given different weights. This analysis helped to finalize the incubator system with the necessary standards and provide valuable insights into its practical application.



## Chapter 3

### **PRESENTATION, ANALYSIS, AND DESIGN RESULT**

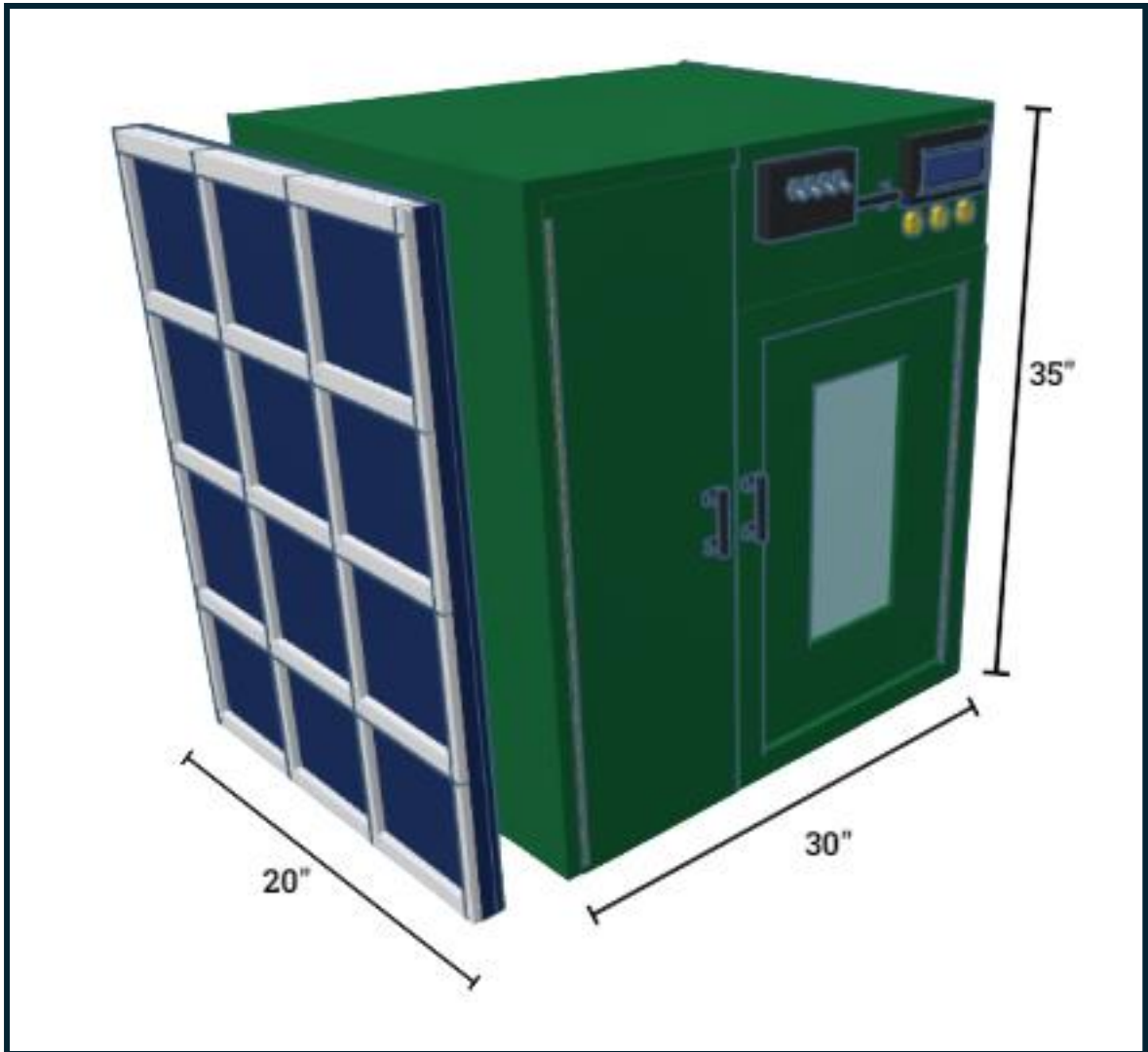
This chapter presents the result of design and development of the system, the presentation, analysis and interpretation of the result of its evaluation.

#### Development of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

The development of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection aims to optimize fertility detection and incubation processes in poultry farming. The core function of the IncuVision system is to integrate computer vision with an automated incubation process, enhancing efficiency by providing accurate fertility detection and incubator monitoring. This approach allows farmers to automate egg candling, reduce manual intervention, and make the incubation process more reliable and efficient.

Figure 6 on the next page showcases the 3D design together with the mount solar panel beside the IncuVision. The 3D model gives a clear understanding of the system's construction and layout, serving as a bridge from design expectations to the final actual device. This visualization provides insight into the design considerations.

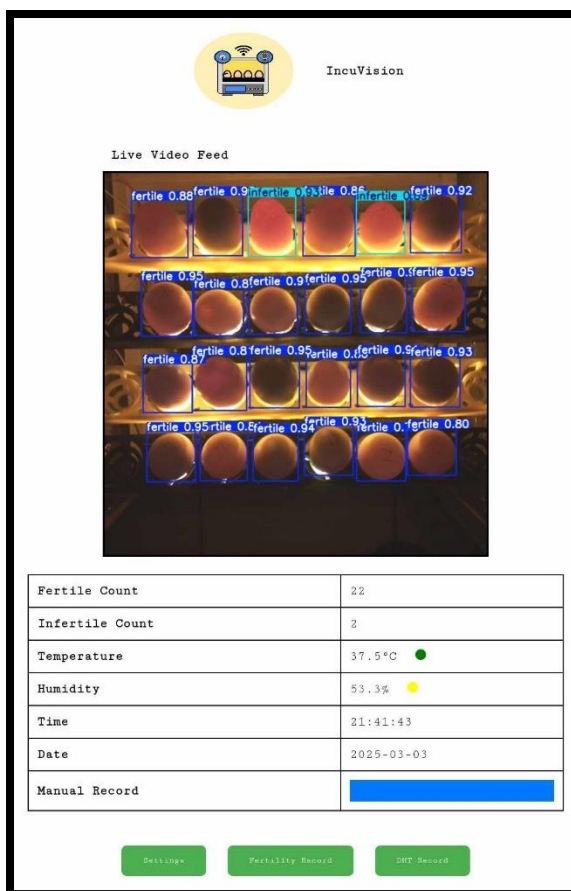
The device utilizes a Raspberry Pi 4B (4GB) as its main control board, along with DHT22 sensors for temperature and humidity monitoring, a Raspberry Pi camera for fertility detection, and an automated egg-tilting mechanism. The system operates in automatic incubation and fertility detection to ensure optimal egg development. Refer to Appendix G for the detailed descriptions of the components in IncuVision.



**Figure 6**

3D Model of IncuVision: An Enhanced Chicken Egg Incubator  
with Computer Vision for Fertility Detection

In the IncuVision system, it utilizes wireless connectivity to enable remote monitoring and control via web-based interface. Wireless connectivity enables remote monitoring and control through a web-based interface, where users can access real-time data on temperature, humidity, and fertility status. The web interface of IncuVision is shown in Figures 7 and 8.



**Figure 7**

### Web-Based Monitoring of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

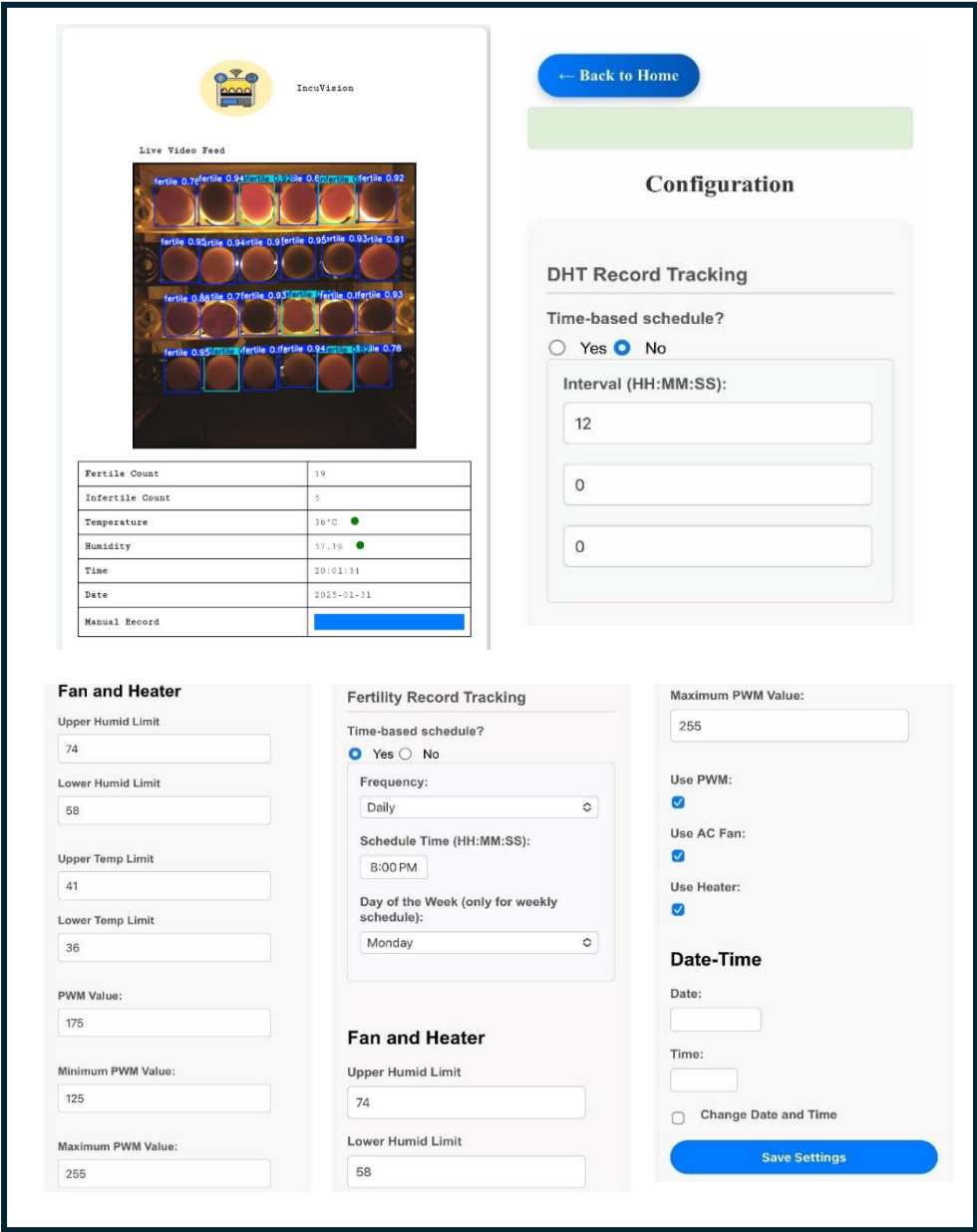
The web interface of IncuVision provides a seamless user experience, as depicted in Figure 7, offers a comprehensive overview of the incubation environment and fertility detection results. The interface prominently displays a live video feed through the Raspberry Pi camera module, enabling real-time monitoring and fertility detection of the chicken eggs. During the candling process, the servo motor-controlled slide cover exposes the camera only when necessary, ensuring optimal operation and preventing unnecessary exposure to heat and moisture. Once the analysis is completed, the fertility records and classification counts, are

automatically stored on the web platform. Mohammed et al. (2023), who discovered that using deep learning-based fertility detection in incubation systems greatly increases classification accuracy and minimizes human error, resulting in higher hatch rates and more dependable automation.

Then, below the video feed, a table provides critical data, including the number of fertile and infertile eggs, temperature, humidity levels, and the date and time. This information is essential for users to assess the incubation conditions and fertility status at a glance. At the bottom of the interface, three buttons—DHT Record, Fertility Record, and Settings—allow users to navigate to specific sections for detailed data review and configuration adjustments. These features ensure that users can easily access and manage the system's functionalities.

Figure 8 displays the user settings configuration section of the web application. This section is designed to provide users with precise control over the incubation environment. It includes options for DHT Record Tracking, where users can enable or disable time-based schedules and set intervals for data recording. The Fan and Heater Controls allow users to define upper and lower limits for humidity and temperature, ensuring optimal conditions for egg incubation. Additionally, PWM (Pulse Width Modulation) settings are available for fine-tuning fan and heater operations. The Fertility Record Tracking section offers scheduling options for fertility monitoring, with choices for daily or weekly tracking and specific time settings. Finally, the Date-Time Configuration section provides a straightforward way to set and adjust the system's date and time, ensuring accurate record-keeping and scheduling. This configuration interface is designed

to be intuitive and user-friendly, enabling efficient management of the incubation process. To access the web app, users can scan a QR code or enter the IP address (192.168.1.22:5000) into their browser, ensuring convenient remote access.



The screenshot displays the IncuVision web application interface, which is divided into several sections for monitoring and controlling an egg incubator.

**Live Video Feed:** A central section showing a grid of 16 egg images. Each image is labeled with a fertility score (e.g., "fertil 0.76", "fertil 0.94"). Below the video feed is a table summarizing the current status:

|                 |            |
|-----------------|------------|
| Fertile Count   | 19         |
| Infertile Count | 5          |
| Temperature     | 36°C       |
| Humidity        | 57.1%      |
| Time            | 20:01:34   |
| Date            | 2025-01-11 |
| Manual Record   |            |

**Configuration:** A section on the right side of the interface for setting up the system. It includes a "Back to Home" button and a "Configuration" header.

**DHT Record Tracking:** A sub-section under Configuration for setting up data logging. It includes a "Time-based schedule?" toggle (set to "No") and an "Interval (HH:MM:SS)" input field (set to "12").

**Fan and Heater:** A section on the left side of the interface for controlling the incubator's environment. It includes input fields for "Upper Humid Limit" (74), "Lower Humid Limit" (58), "Upper Temp Limit" (41), "Lower Temp Limit" (36), "PWM Value" (175), "Minimum PWM Value" (125), and "Maximum PWM Value" (255).

**Fertility Record Tracking:** A section in the middle of the interface for setting up fertility monitoring. It includes a "Time-based schedule?" toggle (set to "Yes"), a "Frequency" dropdown (set to "Daily"), a "Schedule Time (HH:MM:SS)" input field (set to "8:00 PM"), and a "Day of the Week (only for weekly schedule)" dropdown (set to "Monday").

**Date-Time:** A section on the right side of the interface for setting the date and time. It includes input fields for "Date" and "Time", a "Change Date and Time" checkbox, and a "Save Settings" button.

**Figure 8**

User-Friendly Interface of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

The IncuVision system is also equipped with a notification alert system, providing timely updates on egg conditions. Users receive email notifications regarding fertility detection results, including the number of fertile and infertile eggs as shown in Figure 9.



**Figure 9**

#### Notification Alert System of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

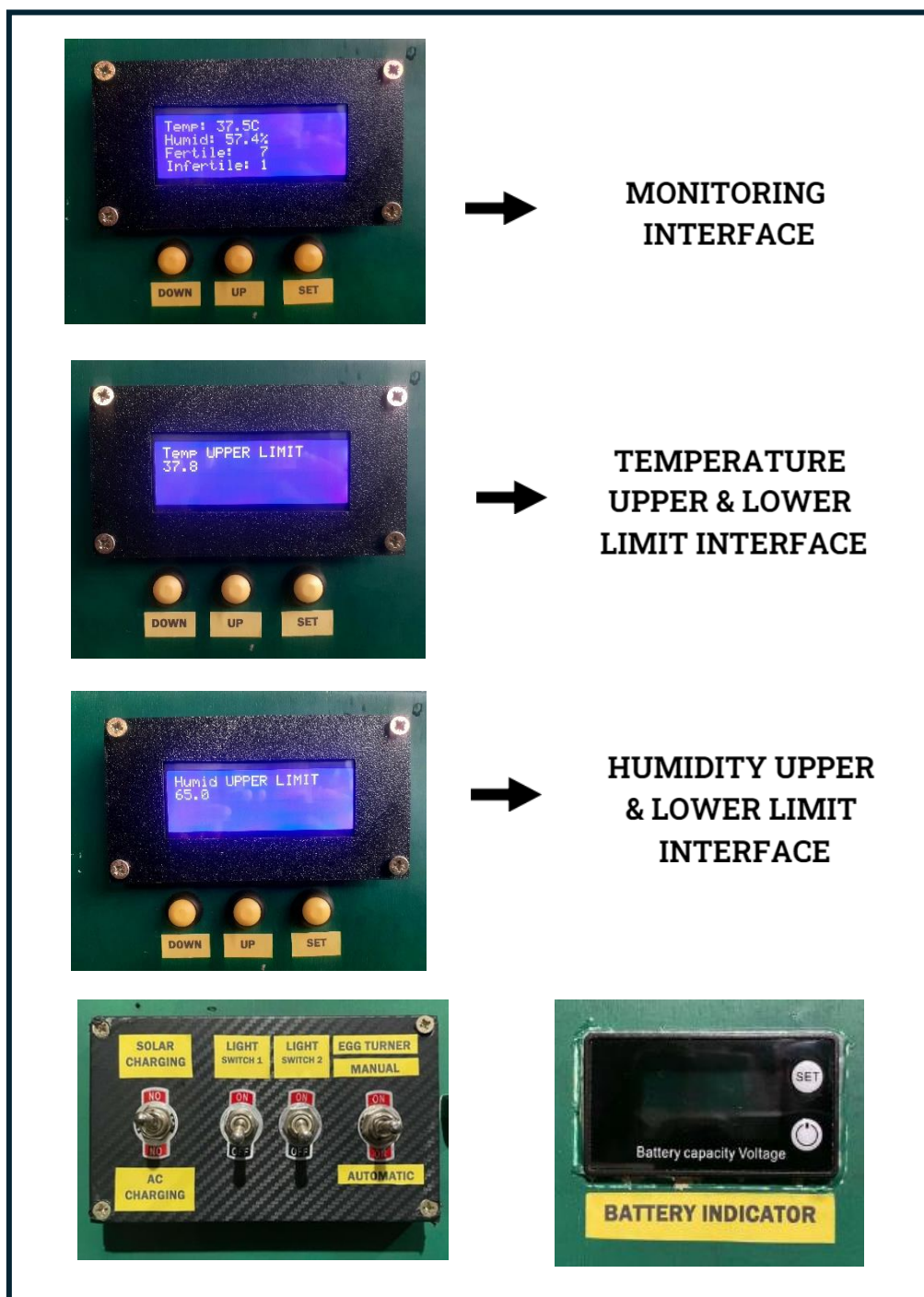
To enhance usability, the IncuVision system integrates an automatic regulation mechanism that allows users to set and control the temperature and humidity according to their desired incubation conditions. The system continuously monitors these parameters and makes adjustments based on user-defined settings, ensuring stable conditions for embryo development. The incubator's automatic temperature regulation system maintains the temperature within the optimal range, which is typically 37.5°C during the setting period and 36.9°C during hatching. A heater and fan mechanism work together to ensure uniform heat distribution, preventing localized overheating or cooling. A DHT22 sensor continuously monitors the temperature, and if deviations occur, the system

automatically activates the heater or fan to restore the set values, reducing the need for manual adjustments.

Maintaining ideal temperature management is critical for improving hatchability and chick quality. Yalcin et al. (2022) underline the need of maintaining an ideal incubation temperature of 37°C to 38°C for good hatchability rates and healthier chicks. Their findings show that even minor variations from this range might have a deleterious impact on embryonic development, resulting in lower hatch success and weaker chicks. This reinforces the necessity of IncuVision's automated temperature regulation system, ensuring a stable incubation environment that enhances embryo development and improves overall hatch rates. Humidity is another critical factor in incubation, as it affects the moisture levels within the eggs, which are essential for proper embryo development and for the membrane that the chicks must break through in order to hatch. The automatic humidity control system enables users to set the desired humidity level, typically within the range of 50 percent to 65 percent, to maintain optimal conditions. A humidity sensor continuously measures the moisture content inside the incubator, ensuring that it remains within the user-defined range. If humidity levels drop below the set threshold, the AC fan is turned on to help release moisture. Conversely, if humidity levels exceed the desired range, the heater reduces excess moisture. There is also another fan that distributes or circulates air evenly inside the incubator.

Aside from the web control and monitoring, the researchers also provided option for users to manually set upper and lower limit of temperature and humidity

using an LCD and buttons, or switch from manual to automatic egg tilting using the switches. The LCD interface, buttons, and switch controls are shown in Figure 10.



**Figure 10**

Manual Option of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection



To address power reliability, the IncuVision incubator incorporates a solar power backup system. It includes a 100W solar panel that ensures continuous operation during power outages, which is particularly beneficial for users in rural or off-grid areas. The detachable solar panel can be mounted on a tripod stand or attached to the incubator for convenience and flexibility. The energy storage system consists of a 12V lead-acid battery with a 100-ampere capacity, which supplies power to the incubator when the primary power source is unavailable. The stored energy ensures that the temperature and humidity control systems, as well as the fertility detection system, remain operational without interruptions.

To regulate power flow and maximize energy efficiency, the system incorporates a 60A Maximum Power Point Tracking (MPPT) charge controller. This charge controller optimizes the energy conversion from the solar panel to the battery, improving charging efficiency and preventing overcharging or deep discharge, which can extend battery lifespan. Additionally, an automated transfer switch (ATS) easily switches between the primary power supply and the solar backup when necessary. The ATS keeps the incubator running without user intervention, lowering the possibility of temperature changes that might hinder embryo growth. Maroma et al. (2023) emphasize the relevance of automated backup power systems in chicken incubation, indicating that continuous energy supply improves hatch rates by ensuring steady incubation conditions. Similarly, Liu et al. (2021) discovered that combining solar power with an ATS system improves energy efficiency and dependability in agricultural automation, ensuring peak performance even in adverse conditions.

For real-time monitoring of battery performance, the system includes a lead-acid battery tester that indicates the battery level and voltage. This tester provides a physical display of the battery's condition, allowing users to check its status at a glance. Unlike a web-based monitoring system, this tester is directly integrated into the incubator, making it a convenient tool for assessing battery health and charge levels this battery indicator tester which is also shown in Figure 9. The system supports simultaneous charging and operation, allowing the battery to be charged while the incubator remains functional. This feature ensures continuous functionality, eliminating the risk of power loss during critical incubation periods.

For the visual representation of the actual device, Figure 11 offers a pictorial representation of the IncuVision. The device operates with a 220V AC with solar power for backup power source which can be mounted on a tripod stand.



**Figure 11**

Prototype of the Chicken Egg Incubator  
with Solar-Powered Backup System

By integrating solar energy storage with a 60A MPPT charge controller and a lead-acid battery tester, the IncuVision incubator enhances its efficiency, reliability, and sustainability. This system provides users with a dependable and energy-efficient solution, ensuring stable incubation conditions even during extended power disruptions

Finally, Figure 12 shows the internal setup of the IncuVision device, including its control panel, egg tray, and power system. It also highlights the main and solar power setup, along with the side-mounted solar panel.



**Figure 12**

Set Up and Internal View of the Developed Device

Overall, the development of IncuVision successfully integrates computer vision and automated incubation, optimizing fertility detection and incubation efficiency. By combining real-time monitoring, automated environmental control, and a solar power backup system, the IncuVision incubator enhances reliability and usability for poultry farmers. To ensure ease of operation, a user manual is provided, as presented in Appendix D, detailing step-by-step instructions for system setup, configuration, and troubleshooting. The implementation of these features ensures a more efficient, accurate, and sustainable incubation process, ultimately reducing manual labor.

#### Test of the Accuracy of Sensor Readings of the Chicken Egg Incubation System

One of the objectives of this study is to evaluate the accuracy of the sensor readings in the chicken egg incubation system, with a specific focus on temperature and humidity levels. To achieve this, the study utilized the DHT22 sensor to measure these parameters inside the incubator. For validation purposes, an analog thermohygrometer was used as the reference standard. To ensure a comprehensive and reliable assessment, temperature and humidity readings were recorded twice daily, specifically at 6:30 AM and 6:30 PM, during the first seven days of the incubation period. These readings were systematically documented in a daily record sheet for analysis.

Table 1 presents the comparison of values from the analog thermohygrometer and the actual temperature measured by the system using a DHT22 sensor. This comparison provides insight into the sensor's accuracy in maintaining optimal incubation conditions.

**Table 1**

Level of Sensor Accuracy of IncuVision Compared with the Analog Thermohygrometer Reference Values in Terms of Temperature

| Day                   | Time      | Reference Temperature (°C) | Measured Temperature (°C) | Mean Accuracy (%) | Interpretation |
|-----------------------|-----------|----------------------------|---------------------------|-------------------|----------------|
| 1                     | 6:30 A.M. | 37.0                       | 37.3                      | 99.46             | Accurate       |
|                       | 6:30 P.M. | 37.5                       | 37.4                      |                   |                |
| 2                     | 6:30 A.M. | 36.5                       | 36.9                      | 99.18             | Accurate       |
|                       | 6:30 P.M. | 37.0                       | 36.8                      |                   |                |
| 3                     | 6:30 A.M. | 37.5                       | 37.8                      | 99.33             | Accurate       |
|                       | 6:30 P.M. | 37.5                       | 37.3                      |                   |                |
| 4                     | 6:30 A.M. | 36.5                       | 37.0                      | 99.18             | Accurate       |
|                       | 6:30 P.M. | 37.5                       | 37.6                      |                   |                |
| 5                     | 6:30 A.M. | 37.0                       | 37.4                      | 99.32             | Accurate       |
|                       | 6:30 P.M. | 37.0                       | 36.9                      |                   |                |
| 6                     | 6:30 A.M. | 37.5                       | 37.7                      | 99.60             | Accurate       |
|                       | 6:30 P.M. | 37.5                       | 37.6                      |                   |                |
| 7                     | 6:30 A.M. | 37.0                       | 37.4                      | 99.19             | Accurate       |
|                       | 6:30 P.M. | 37.0                       | 37.2                      |                   |                |
| Overall Mean Accuracy |           |                            |                           | 99.32             | Accurate       |

As shown in Table 1, the recorded mean accuracy percentages for the system's temperature measurements range from 99.18 percent to 99.60 percent over the first seven days of incubation, which are within the acceptable range. As a result, the system achieved an overall mean accuracy of 99.32 percent, confirming its reliability in temperature measurements. The accuracy of the system was evaluated by comparing the temperature readings of the DHT22 sensor

against reference temperatures obtained from an analog thermohygrometer. According to the DHT22 sensor's datasheet, its temperature accuracy is specified as  $\pm 0.5^{\circ}\text{C}$ . When this margin of error is considered, the corresponding minimum accuracy of the sensor is approximately 98.67 percent, based on the obtained reference temperatures. Given that all computed accuracy values exceed this threshold, the temperature measurement system implemented in IncuVision demonstrates a high level of precision. This ensures that the incubation environment remains stable, thereby supporting the critical conditions required for successful embryo development.

This accuracy level aligns with research findings emphasizing the importance of stable temperature conditions during incubation. Studies by Yalcin et al. (2022) and De Almeida et al. (2016) highlight that maintaining the correct temperature is essential for optimal hatchability and proper embryo development. In particular, research has shown that even minor temperature fluctuations can significantly impact embryonic growth, hatchability rates, and chick quality. The results confirm that IncuVision effectively provides the necessary thermal conditions for successful incubation, reinforcing the importance of precise temperature regulation in poultry production.

Following this, Table 2 presents the comparison of values from the analog thermohygrometer and the actual humidity measured by the system using a DHT22 sensor during the first seven days of incubation. The subsequent analysis examines the system's capability in maintaining accurate humidity levels, further validating its reliability as a fully integrated incubation solution.

**Table 2**

Level of Sensor Accuracy of IncuVision Compared with the Analog Thermohygrometer Reference Values in Terms of Humidity

| Day                   | Time      | Reference Humidity (%) | Measured Humidity (%) | Mean Accuracy (%) | Interpretation |
|-----------------------|-----------|------------------------|-----------------------|-------------------|----------------|
| 1                     | 6:30 A.M. | 57.0                   | 58.7                  | 96.51             | Accurate       |
|                       | 6:30 P.M. | 57.5                   | 59.8                  |                   |                |
| 2                     | 6:30 A.M. | 61.0                   | 63.0                  | 96.11             | Accurate       |
|                       | 6:30 P.M. | 60.0                   | 62.7                  |                   |                |
| 3                     | 6:30 A.M. | 62.0                   | 63.1                  | 98.69             | Accurate       |
|                       | 6:30 P.M. | 58.5                   | 59.0                  |                   |                |
| 4                     | 6:30 A.M. | 60.0                   | 62.4                  | 96.22             | Accurate       |
|                       | 6:30 P.M. | 59.0                   | 61.1                  |                   |                |
| 5                     | 6:30 A.M. | 62.0                   | 63.2                  | 97.24             | Accurate       |
|                       | 6:30 P.M. | 64.0                   | 61.7                  |                   |                |
| 6                     | 6:30 A.M. | 61.0                   | 59.9                  | 96.74             | Accurate       |
|                       | 6:30 P.M. | 61.5                   | 58.6                  |                   |                |
| 7                     | 6:30 A.M. | 63.0                   | 64.4                  | 98.21             | Accurate       |
|                       | 6:30 P.M. | 59.0                   | 59.8                  |                   |                |
| Overall Mean Accuracy |           |                        |                       | 97.10             | Accurate       |

As shown in Table 2, the computed mean accuracy percentages for the system's humidity measurements range from 96.11 percent to 98.69 percent over the first seven days of incubation, indicating that the system consistently operates within the expected tolerance range. As a result, the system achieved an overall mean accuracy of 97.10 percent, confirming its reliability in humidity measurements. Based on the DHT22 sensor's datasheet, its humidity accuracy is specified as  $\pm 2$  percent to 5 percent. When this margin of error is considered, the

recorded humidity accuracy values remain well within the expected range. The consistency of these readings suggests that the system effectively regulates humidity levels, minimizing fluctuations that could impact embryo development. Maintaining appropriate humidity levels is crucial during incubation, as improper humidity regulation can lead to excessive moisture loss or retention within the egg, affecting hatchability rates. The obtained results confirm that IncuVision provides stable and accurate humidity conditions, ensuring optimal embryonic development.

Additionally, this result aligns with findings by Oliveira et al. (2020) and Dutta and Anjum (2021) as they noted that improper humidity control can negatively impact the incubation process, leading to reduced hatchability and chick quality. By ensuring stable humidity conditions, IncuVision effectively minimizes the risks associated with excessive moisture loss or retention, supporting a more consistent and successful hatching process. Therefore, the findings affirm the system's capability to maintain ideal humidity levels essential for optimal incubation outcomes.

#### Determine the Percent Error of the Automated Chicken Egg Fertility Detection System with Reference to Traditional Candling

The fertility detection capability of the IncuVision system was assessed through percent error analysis, which compares the system's automated classifications with those obtained using traditional candling. This evaluation aims to determine the accuracy of the system in distinguishing between fertile and infertile eggs.

Table 3 presents the percent error in the classification of fertile eggs,



providing a comprehensive assessment of the system's performance relative to traditional candling.

**Table 3**

Computed Percent Error of the Automated Chicken Egg Fertility Detection System with Reference to Traditional Candling

| Day | FERTILE               |                    |                   | INFERTILE             |                    |                   |
|-----|-----------------------|--------------------|-------------------|-----------------------|--------------------|-------------------|
|     | Traditional Candling  | Automatic Candling | Percent Error (%) | Traditional Candling  | Automatic Candling | Percent Error (%) |
| 1   | 0                     | 0                  | 0.00              | 24                    | 24                 | 0.00              |
| 2   | 0                     | 0                  | 0.00              | 24                    | 24                 | 0.00              |
| 3   | 0                     | 0                  | 0.00              | 24                    | 24                 | 0.00              |
| 4   | 0                     | 0                  | 0.00              | 24                    | 24                 | 0.00              |
| 5   | 9                     | 13                 | 44.44             | 15                    | 11                 | 26.67             |
| 6   | 17                    | 15                 | 11.76             | 7                     | 9                  | 28.57             |
| 7   | 20                    | 22                 | 10.00             | 4                     | 2                  | 50.00             |
|     | Overall Percent Error |                    | 9.46              | Overall Percent Error |                    | 15.03             |

The data presented in Table 3 shows the percentage error of the IncuVision system's automated fertility detection compared to traditional candling over the first seven days of incubation. The analysis includes the classification errors for both fertile and infertile eggs. During the first four days, both methods yielded identical results, leading to a 0 percent error rate for both fertile and infertile eggs. However, as incubation progressed, discrepancies emerged as the system exhibited higher error rates. This suggests that the system may struggle with detecting early fertility

indicators due to limited embryonic development visibility. Overall, the computed percent error of 9.46 percent and 15.03 percent indicates that the IncuVision system achieved an accuracy of 90.54 percent for fertile eggs and 84.97 percent for infertile eggs, demonstrating its potential for reliable fertility classification.

These findings align with the study of Tolentino et al. (2022), which examined an automatic incubator using a Region-based Convolutional Neural Network (R-CNN) for detecting the maturity of chicken and duck eggs, achieving an accuracy of 80.5 percent. Their study highlighted those challenges in fertility detection often stem from low image quality and scattered candlelight, which affect embryo visibility. Based on that study, an accuracy threshold of 80 percent was set to ensure the deep learning model's effectiveness. Given this standard, the IncuVision system meets the required accuracy for effective fertility classification, reinforcing its reliability for automated incubation systems. However, to further reduce misclassification errors, enhancements in algorithm optimization and image processing techniques should be explored. Refinements in these areas could improve the system's overall consistency and accuracy.

#### The Level of Acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

The researchers were able to gather the acceptability of the project according to functional suitability, performance efficiency, usability, reliability, and maintainability.

This assessment provides insight into how well the system meets its intended objectives. The following tables present the computed level of acceptability of the study's goals.

Table 4 displays the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in terms of functional suitability.

**Table 4**

Computed Weighted Mean of the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in Terms of Functional Suitability

| Functional Suitability                                                                                                                                                   | Weighted Mean | Verbal Interpretation |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|
| 1.1 The developed system maintains optimal incubation conditions (e.g., temperature, humidity) throughout the incubation process.                                        | 4.84          | Very Much Acceptable  |
| 1.2 The YOLO deep learning model consistently and accurately detects the fertility status of chicken eggs.                                                               | 4.75          | Very Much Acceptable  |
| 1.3 The developed system monitors and logs data such as temperature, humidity, and number of fertile and infertile chicken eggs, accessible through the web application. | 4.94          | Very Much Acceptable  |
| 1.4 The developed system provides timely notifications or alerts for detected abnormal incubation conditions or infertile eggs.                                          | 4.91          | Very Much Acceptable  |
| 1.5 The incubator automatically switches to solar power during power interruptions and seamlessly maintains functionality without disrupting operations.                 | 4.97          | Very Much Acceptable  |
| Average Weighted Mean                                                                                                                                                    | 4.88          | Very Much Acceptable  |

As shown in Table 4, Item 1.5, which evaluates the system's ability to automatically switch to solar power during power interruptions while maintaining seamless functionality, received the highest score with a weighted mean of 4.97, interpreted as Very Much Acceptable. On the other hand, item 1.2, assessing the

consistency and accuracy of the YOLO deep learning model in detecting the fertility status of chicken eggs, obtained the lowest score with a weighted mean of 4.75, though still interpreted as Very Much Acceptable. The average weighted mean of 4.88 is interpreted as Very Much Acceptable, indicating a high level of acceptance, suggesting that the system effectively meets the functional suitability requirements for optimal incubation and fertility detection.

The system's ability to maintain stable temperature and humidity control aligns with the findings of Smith et al. (2020), who emphasized that precise regulation of these factors significantly improves hatch rates and reduces embryonic mortality. Their research supports the importance of maintaining an optimal incubation environment, reinforcing the effectiveness of IncuVision in ensuring the successful development of embryos. Furthermore, the system's automated fertility detection capability supports the study of Vargas et al. (2018), which highlighted the inefficiency of traditional egg candling methods. Manual candling is not only time-consuming but also prone to human error, making automated solutions like IncuVision highly advantageous in modern hatchery operations. By integrating advanced technologies, IncuVision demonstrates its potential to enhance incubation practices, improve efficiency, and contribute to better hatchery management in poultry production.

Table 5 shows the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in terms of performance efficiency.

**Table 5**

Computed Weighted Mean of the IncuVision: An Enhanced Chicken Egg  
Incubator with Computer Vision for Fertility Detection  
in Terms of Performance Efficiency

| Performance Efficiency                                                                                                                                             | Weighted Mean | Verbal Interpretation |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|
| 2.1 The incubator powers on and initializes its components without significant delays.                                                                             | 4.72          | Very Much Acceptable  |
| 2.2 Sensors quickly provide readings of incubation conditions, including temperature and humidity.                                                                 | 4.78          | Very Much Acceptable  |
| 2.3 The YOLO model for fertility detection processes results promptly to minimize waiting times for users.                                                         | 4.78          | Very Much Acceptable  |
| 2.4 The incubator can handle multiple processes simultaneously, such as fertility detection and incubation condition adjustments, without performance degradation. | 4.84          | Very Much Acceptable  |
| 2.5 The incubator operates efficiently without generating excessive heat or noise.                                                                                 | 4.97          | Very Much Acceptable  |
| Average Weighted Mean                                                                                                                                              | 4.82          | Very Much Acceptable  |

As shown in Table 5, Item 2.5, which evaluates the system's ability to operate efficiently without generating excessive heat or noise, received the highest score with a weighted mean of 4.97, interpreted as Very Much Acceptable. Meanwhile, item 2.1, assessing the incubator's ability to power on and initialize its components without significant delays, obtained the lowest score with a weighted mean of 4.72, still interpreted as Very Much Acceptable. The average weighted mean of 4.82, interpreted as Very Much Acceptable, indicates a high level of acceptance, suggesting that the system effectively optimizes incubation processes while maintaining efficiency. The system's performance efficiency is consistent with the findings of Niranjana et al. (2021), who developed a smartphone-based IoT-

integrated incubator for monitoring temperature, humidity, and egg turning. Their study demonstrated that automation significantly enhanced incubation control, reducing manual labor while ensuring optimal hatching conditions.

Table 7 presents the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in terms of usability.

**Table 6**

Computed Weighted Mean of the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in Terms of Usability

| Usability                                                                                                                                                                  | Weighted Mean | Verbal Interpretation |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|
| 3.1 The incubator is simple to assemble and configure, even for users with minimal technical expertise.                                                                    | 4.72          | Very Much Acceptable  |
| 3.2 The temperature and humidity sensor readings are presented in a clear and understandable manner.                                                                       | 4.84          | Very Much Acceptable  |
| 3.3 Fertility status for each egg is displayed clearly with easily distinguishable markers, such as color-coded indicators or percentage probabilities, to guide the user. | 4.97          | Very Much Acceptable  |
| 3.4 Users can conveniently access, monitor, and control incubation settings through physical controls or the web-based application.                                        | 4.94          | Very Much Acceptable  |
| 3.5 The web-based application provides real-time updates on temperature, humidity, and fertility detection results, ensuring users always have the latest information.     | 4.97          | Very Much Acceptable  |
| Average Weighted Mean                                                                                                                                                      | 4.89          | Very Much Acceptable  |

As presented in Table 6, Item 3.3, which evaluates the clarity of fertility status display through easily distinguishable markers such as color-coded indicators or percentage probabilities, received the highest score with a weighted mean of 4.97, interpreted as Very Much Acceptable. Similarly, Item 3.5, assessing the web-based application's ability to provide real-time updates on temperature, humidity, and fertility detection results, also obtained a weighted mean of 4.97, highlighting its effectiveness in ensuring users always have the latest information.

On the other hand, Item 3.1, which measures the ease of assembling and configuring the incubator for users with minimal technical expertise, received the lowest score with a weighted mean of 4.72, though still interpreted as Very Much Acceptable. The average weighted mean of 4.89, interpreted as Very Much Acceptable, indicates a high level of usability, suggesting that the system is user-friendly and accessible, even for individuals with limited technical knowledge.

The system's usability aligns with the findings of Kim et al. (2020), who emphasized that intuitive user interfaces and real-time monitoring capabilities significantly enhance the efficiency, accessibility, and overall user experience of automated incubation systems. By integrating a user-friendly design with automated monitoring, the system minimizes the need for manual intervention, making it more convenient and efficient for users of varying technical expertise.

Additionally, the use of clear fertility indicators supports the recommendations of Garcia et al. (2019), who found that visual markers, such as color-coded signals or percentage-based probabilities, improve user decision-making and reduce errors in fertility assessment. Providing easily interpretable

fertility status updates ensures that users can make timely and informed decisions, ultimately contributing to better incubation outcomes.

Table 7 displays the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in terms of reliability.

**Table 7**

Computed Weighted Mean of the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in Terms of Reliability

| Reliability                                                                                                                                                          | Weighted Mean | Verbal Interpretation |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|
| 4.1 Temperature and humidity sensors deliver accurate readings.                                                                                                      | 4.84          | Very Much Acceptable  |
| 4.2 Key components of the incubator (e.g., heaters, fans) respond effectively to maintain optimal incubation conditions.                                             | 4.84          | Very Much Acceptable  |
| 4.3 The YOLO deep learning model consistently identifies fertility status with high accuracy, minimizing false positives and false negatives across repeated trials. | 4.91          | Very Much Acceptable  |
| 4.4 The YOLO model processes and delivers fertility detection results quickly, minimizing delays in decision-making during incubation.                               | 4.88          | Very Much Acceptable  |
| 4.5 The incubator operates continuously and reliably over extended periods, with no significant failures or downtime.                                                | 4.88          | Very Much Acceptable  |
| Average Weighted Mean                                                                                                                                                | 4.87          | Very Much Acceptable  |

As presented in Table 7, Item 4.3, which evaluates the accuracy of the YOLO deep learning model in identifying fertility status while minimizing false positives and negatives, received the highest score with a weighted mean of 4.91,



interpreted as Very Much Acceptable. This highlights the system's effectiveness in accurately distinguishing fertile and infertile eggs, which is crucial for optimizing hatch rates and reducing resource wastage. Meanwhile, Items 4.1 and 4.2, assessing the accuracy of temperature and humidity sensor readings and the responsiveness of key incubator components, obtained the lowest scores with weighted means of 4.84, though still interpreted as Very Much Acceptable. The consistently high ratings suggest that the system maintains precise environmental conditions essential for successful embryo development. The average weighted mean of 4.87 is also interpreted as Very Much Acceptable, indicating a high level of reliability, ensuring consistent performance across various incubation conditions.

The system's reliability aligns with the findings of Patel et al. (2021), who emphasized that accurate sensor readings and automated environmental control mechanisms significantly enhance incubation success rates. By maintaining stable environmental conditions, the system minimizes the risks associated with manual adjustments and external fluctuations, thereby increasing the likelihood of successful hatching. Additionally, the YOLO model's high accuracy and quick processing time support the findings of Zhang et al. (2020), who demonstrated that deep learning-based fertility detection reduces misclassification errors and improves incubation efficiency.

Table 8 shows the computed weighted mean, verbal interpretation, and average weighted mean of the level of acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in terms of

maintainability.

**Table 8**

Computed Weighted Mean of the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection in Terms of Maintainability

| Maintainability                                                                                                            | Weighted Mean | Verbal Interpretation |
|----------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------|
| 5.1 The incubator's design facilitates easy cleaning without requiring disassembly of key components.                      | 4.91          | Very Much Acceptable  |
| 5.2 Internal wiring and connections are well-organized for easy inspection and maintenance.                                | 4.75          | Very Much Acceptable  |
| 5.3 Replacement of commonly used parts (e.g., fans, heaters, sensors) is user-friendly, with components easily accessible. | 4.97          | Very Much Acceptable  |
| 5.4 Components are labeled with detailed information, ensuring easy identification for repairs or replacements.            | 4.97          | Very Much Acceptable  |
| 5.5 Maintenance tasks can be carried out using standard tools, eliminating the need for specialized equipment.             | 4.94          | Very Much Acceptable  |
| Average Weighted Mean                                                                                                      | 4.91          | Very Much Acceptable  |

As shown in Table 8, Items 5.3 and 5.4, which evaluate the ease of replacing commonly used components such as fans, heaters, and sensors, as well as the clarity of component labeling for easy identification during repairs, received the highest scores with weighted means of 4.97, interpreted as Very Much Acceptable. Meanwhile, Item 5.2, which assesses the organization of internal wiring and connections for easy inspection and maintenance, obtained the lowest score with a weighted mean of 4.75, but still interpreted as Very Much Acceptable. The average weighted mean of 4.91, interpreted as Very Much Acceptable, indicates a high level of maintainability, suggesting that the system is designed for

efficient upkeep, ensuring long-term usability and minimal downtime.

The system's maintainability aligns with the findings of Bansal et al. (2015), who emphasized that a well-maintained fan and heater ensure stable and reliable temperature control, reducing the need for frequent repairs. Their study highlights the significance of organized and properly labeled components in optimizing system performance and minimizing downtime, which supports the high acceptability scores observed in this study. Furthermore, ensuring easy access to key components simplifies troubleshooting and routine maintenance. This enhances the overall longevity and efficiency of the incubator, making it a cost-effective and sustainable solution.

Table 9 presents the summary of the level of acceptability of the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection. It is very acceptable in terms of functional suitability, performance efficiency, usability, reliability, and maintainability.

**Table 9**

Composite Table on the Level of Acceptability of the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

| <b>Variables</b>                     | <b>Mean</b> | <b>Verbal Interpretation</b> |
|--------------------------------------|-------------|------------------------------|
| Functional Suitability               | 4.88        | Very Much Acceptable         |
| Performance Efficiency               | 4.82        | Very Much Acceptable         |
| Usability                            | 4.89        | Very Much Acceptable         |
| Reliability                          | 4.87        | Very Much Acceptable         |
| Maintainability                      | 4.91        | Very Much Acceptable         |
| <b>Overall Average Weighted Mean</b> | <b>4.87</b> | <b>Very Much Acceptable</b>  |

Table 9 shows that IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection is highly acceptable in terms of functional suitability, performance efficiency, usability, reliability, and maintainability. With an overall weighted mean of 4.87, interpreted as Very Much Acceptable, the system effectively automates and optimizes the incubation process. The highest-rated aspect, maintainability (4.91), highlights its user-friendly design, ensuring easy maintenance and long-term usability. Usability (4.89) and functional suitability (4.88) also received high scores, reinforcing the system's efficiency in fertility detection and incubation monitoring. Although performance efficiency (4.82) ranked the lowest, it remains at the highest acceptability level, ensuring optimal operation without significant inefficiencies. Overall, the findings confirm that the developed incubator meets its objectives, providing a reliable and efficient solution for poultry incubation through computer vision and automation.

## Chapter 4

### **SUMMARY OF DESIGN RESULTS, CONCLUSIONS AND RECOMMENDATIONS**

This chapter presents the summary of design results, conclusions, and recommendations of the study.

#### Summary of Design Results

The discussion of the design results is summarized below.

1. The Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection was developed to optimize the candling and incubation processes in poultry farming. The system utilizes a Raspberry Pi 4B as the central control unit, equipped with a Raspberry Pi camera that provides an internal view of the incubator, allowing the YOLOv8 deep learning model to classify eggs as fertile or infertile during the candling process. The system automatically regulates temperature and humidity using a DHT22 sensor, with status updates displayed on an LCD and accessible via a web-based interface for remote monitoring. Users receive email notifications regarding fertility detection. The incubator also features manual controls via LCD, buttons, and switches for adjusting settings. To ensure continuous operation, the incubator includes a solar-powered backup with a 100W panel, a 12V lead-acid battery, and an MPPT charge controller. By integrating automation, computer vision, and renewable energy, the system enhances reliability, precision, and sustainability in poultry incubation.
2. The IncuVision system was tested for accuracy in temperature control, humidity control, and fertility detection, demonstrating high performance across all

- objectives. For temperature measurements, the DHT22 sensor achieved mean accuracy ranging from 99.18 percent to 99.60 percent, exceeding the sensor's minimum expected accuracy of 98.67 percent based on its plus or minus 0.5 degrees Celsius tolerance, ensuring optimal conditions for embryo development. The system maintained stable temperatures throughout the incubation period, reducing fluctuations that could negatively impact hatchability. Humidity control also performed reliably, with mean accuracy between 96.11 percent and 98.69 percent, well within the DHT22 sensor's plus or minus 5 percent tolerance, maintaining appropriate moisture levels for hatchability. The system's ability to automatically adjust humidity levels based on sensor readings ensured that optimal moisture conditions were maintained.
3. The accuracy of IncuVision's automated fertility detection was evaluated by comparing its results to traditional candling using percent error analysis. Data collected over the first seven days of incubation showed that both methods had identical classifications in the first four days, resulting in zero percent error. As incubation progressed, discrepancies emerged. But overall, the system obtained a percent error of 9.46 percent and 15.03 percent, which indicated that the IncuVision system achieved an accuracy of 90.54 percent for fertile eggs and 84.97 percent for infertile eggs. This result exceeded the set accuracy threshold of 80 percent.
  4. A survey was conducted with 32 participants, comprising 7 experts and 25 potential end-users, to determine the level of acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Predictive Fertility

Detection. The findings revealed a strong level of approval across various criteria. Functional suitability received an average weighted mean of 4.88, which was verbally interpreted as very much acceptable. In terms of performance efficiency, the system achieved a weighted mean of 4.82, also interpreted as very much acceptable. Usability was rated with a weighted mean of 4.89, verbally interpreted as very much acceptable. For reliability, the device scored a mean of 4.87, interpreted as very much acceptable. Lastly, for maintainability, the device received a mean of 4.91, also interpreted as very much acceptable. The overall average weighted mean of 4.87 further indicated that the developed device was very much acceptable across all variables.

### Conclusions

The following conclusions were drawn based on the aforementioned findings of the developed IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection.

1. The development of IncuVision system successfully optimized the candling and incubation processes in poultry farming. By integrating automation, computer vision, and renewable energy, the system demonstrated enhanced precision, efficiency, and sustainability. The combination of a Raspberry Pi 4B, YOLOv8 deep learning model, and real-time monitoring capabilities ensured accurate fertility detection and stable incubation conditions.
2. The IncuVision system was proved to be a highly reliable solution for automated egg incubation, with its sensor readings demonstrating exceptional accuracy in both temperature and humidity control. The DHT22 sensor

- exceeded its expected accuracy thresholds, ensuring optimal conditions for embryo development and hatchability. The results confirm that IncuVision successfully integrates advanced sensor technology to optimize poultry farming, offering a sustainable and efficient incubation solution.
3. The percent error analysis confirmed the reliability and effectiveness of IncuVision in automated fertility detection. With low percent errors and high accuracy rates, the system successfully met and exceeded the established accuracy threshold. Despite minor misclassifications, the system consistently identified egg fertility, proving IncuVision to be a highly accurate and reliable solution for automated incubation systems.
  4. The acceptability evaluation of IncuVision based on ISO/IEC 25010 showed a strong level of approval among 32 participants, including 7 experts and 25 potential end-users. The system received very much acceptable ratings across all criteria. The functional suitability, performance efficiency, usability, reliability, and maintainability, with ratings ranging from 4.87 to 4.91. Overall, these results confirm the system's effectiveness, reliability, and strong user acceptance in poultry farming.

### Recommendations

Based on the results of the study, the researchers arrived on the following recommendations:

1. Improve ventilation and sealing to prevent air leaks, ensuring stable humidity and optimal incubation conditions.
2. Enhance AI model training with a larger, diverse dataset to improve fertility



detection accuracy. Further testing and optimization of YOLOv8 can reduce misclassifications.

3. Increase egg capacity for greater efficiency and scalability, making the system suitable for large-scale poultry farming.
4. Upgrade the processing unit to enhance system performance, reduce boot time, and improve real-time image processing and web-based monitoring.
5. Use darker or non-reflective egg trays to minimize light reflection, reducing camera interference and improving fertility detection accuracy.

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# **APPENDICES**

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## **Appendix B**

### Appointment Letters



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### College of Engineering - URS Morong

## APPOINTMENT

August 30, 2024

**ENGR. JENADEL DL. ANTIPOLLO**  
Faculty  
This College

Dear Madame;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Research Adviser**.

The following are the responsibilities of the Research Adviser:

1. Supervises and monitors the research work of the advisees upon accepting the assignment as an adviser;
2. Sees that thesis meets the requirements of research before endorsing to the members of the panel and research instructor for consultation;
3. Certifies to the readiness of the advisees to render final defense.
4. Requires the advisees to give a copy of the manuscript to the panel members one week before the scheduled oral defense
5. Works closely with the advisees and monitors the revision of the manuscript considering the suggestions of the panel and certifies the correctness of the same before referring to the members of the panel and research instructors;
6. Coordinates with the College Research Coordinator on matters related to research activities; and
7. Undertakes the function of the recorder who records proceedings of the defense and supplies the advisees with the summarized recommendations / suggestions of the panel.

The students who shall be under your supervision include:

| <i>Names</i>                 | <i>Course</i> |
|------------------------------|---------------|
| Ashanty Madeline DC. Cruz    | BS ECE        |
| Brian Ezekiel D. Batalon     | BS ECE        |
| Tristan Manuel T. Sto. Tomas | BS ECE        |
| Mark Jayson F. Villacura     | BS ECE        |

The undergraduate thesis proposal is entitled **INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION**

Thank you for your usual support to the research program of the College.

Respectfully yours,

  
**NORMAN C. FRANCISCO, REE**  
College Research Coordinator

Conforme:

  
**JENADEL DL. ANTIPOLLO, ECE**  
Research Adviser

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Tel. 8539-9940 to 41  
Tel. 8539-9950 to 56

URS Piliña  
URS Rodriguez  
URS Tawtaw

Tel. 8539-9942 to 44  
Tel. 8539-9945 to 47  
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**College of Engineering - URS Morong**

**APPOINTMENT**

August 30, 2024

**ENGR. RIO CAMILLE M. PEDROCILLO**

Faculty  
This College

Dear Madame;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Panel Chairperson**.

The following are the responsibilities of the Panel Chairperson:

1. Acts as presiding officer;
2. Decides on the procedure of the defense;
3. Settles disagreements among the panel members and/or comments and suggestions;
4. Solicits the decision of the panel, and
5. Consolidates and announces the rating.

The students who shall be under your scrutiny include:

| <i>Names</i>                 | <i>Course</i> |
|------------------------------|---------------|
| Ashanty Madeline DC. Cruz    | BS ECE        |
| Brian Ezekiel D. Batalon     | BS ECE        |
| Tristan Manuel T. Sto. Tomas | BS ECE        |
| Mark Jayson F. Villacura     | BS ECE        |

The undergraduate thesis proposal is entitled **INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION**

Thank you for your usual support to the research program of the College.

Respectfully yours,

  
**NORMAN C. FRANCISCO, REE**  
College Research Coordinator

Conforme:

  
**RIO CAMILLE M. PEDROCILLO, ECE**  
Panel Chairperson

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URS Rodriguez  
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**College of Engineering - URS Morong**

**APPOINTMENT**

August 30, 2024

**ENGR. CRYSTALEENE JADE A. SANTOS**

Faculty  
This College

Dear Madame;

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as **Research Expert**.

The following are the responsibilities of the Thesis Expert:

1. Provide subject matter expertise for the study;
2. Suggests appropriate statistical treatment for the study;
3. Guides the researchers in computation of data;
4. Guides the researchers in the correct analysis and interpretation of data.

The students who shall be under your scrutiny include:

| <i>Names</i>                 | <i>Course</i> |
|------------------------------|---------------|
| Ashanty Madeline DC. Cruz    | BS ECE        |
| Brian Ezekiel D. Batalon     | BS ECE        |
| Tristan Manuel T. Sto. Tomas | BS ECE        |
| Mark Jayson F. Villacura     | BS ECE        |

The undergraduate thesis proposal is entitled **INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION**

Thank you for your usual support to the research program of the College.

Respectfully yours

  
**NORMAN C. FRANCISCO, REE**  
College Research Coordinator

Conforme:

  
**CRYSTALEENE JADE A. SANTOS, CpE**  
Research Expert

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| URS Antipolo   | Tel. 8539-9932 to 34 | URS Cardona | Tel. 8539-9940 to 41 | URS Rodriguez | Tel. 8539-9945 to 47 |
| URS Binangonan | Tel. 8539-9935 to 37 | URS Morong  | Tel. 8539-9950 to 56 | URS Tautav    | Tel. 8539-9948 to 49 |



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**College of Engineering - URS Morong**  
**APPOINTMENT**

August 30, 2024

**ENGR. ERICKSON T. MARCOS**

Faculty  
This College

Dear Sir,

In consideration of your qualifications in the field of research, the College Research Council represented by the undersigned, upon the recommendations of the Research Professor has approved your appointment as *Critic Reader*.

The following are the responsibilities of the Critic Reader:

1. Reviews all the grammatical aspects;
2. Certifies that the manuscript is already free of grammatical errors and is ready for submission to the office of the Dean of the College;

The students who shall be under your scrutiny include:

| <i>Names</i>                 | <i>Course</i> |
|------------------------------|---------------|
| Ashanty Madeline DC. Cruz    | BS ECE        |
| Brian Ezekiel D. Batalon     | BS ECE        |
| Tristan Manuel T. Sto. Tomas | BS ECE        |
| Mark Jayson F. Villacura     | BS ECE        |

The undergraduate thesis proposal is entitled **INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION**

Thank you for your usual support to the research program of the College.

Respectfully yours,

  
**NORMAN C. FRANCISCO, REE**  
College Research Coordinator

Conforme:

  
**ERICKSON T. MARCOS, MSc**  
Critic Reader

---

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**College of Engineering - URS Morong**

**DESIGNATION OF UNDERGRADUATE THESIS ADVISORY/ ORAL EXAMINATION  
COMMITTEE**

Date: September 12, 2024

TO: Engr. Rio Camille M. Pedrocillo  
Engr. Crystaleene Jade A. Santos  
Engr. Erickson T. Marcos

We have the honor to inform you that you are hereby requested as Adviser, Chairperson, Critic Reader and Expert for Ashanty Madeline DC. Cruz, Brian Ezekiel D. Batalon, Tristan Manuel T. Sto. Tomas, and Mark Jayson F. Villacura students of **BS Electronics Engineering** as follows:

Panel Chairman: Rio Camille M. Pedrocillo, ECE  
Expert: Crystaleene Jade A. Santos, CpE  
Critic Reader: Erickson T. Marcos, MSc

CONFORME

[Signature]  
[Signature]  
[Signature]

The approved thesis title is INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION, an outline of which is herewith attached for your perusal, evaluation and approval.

As such, you are humbly asked to advise and guide the advisee during the operation conduct of the thesis to ensure high scholastic standards of the research.

Thank you very much.

Very truly yours,

[Signature]  
Jenadel DL. Antipolo, ECE  
Research Adviser, Research Instructor,  
Electronics Engineering

Noted by:

[Signature]  
Michael L. Pascua, MSCM  
Dean, College of Engineering

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## **Appendix C**

### Communication Letters and Forms



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Bachelor of Science in Electronics Engineering



**RESEARCHER-MADE QUESTIONNAIRE CHECKLIST FOR THE  
UNDERGRADUATE THESIS INCUVISION: AN ENHANCED  
CHICKEN EGG INCUBATOR WITH COMPUTER VISION  
FOR FERTILITY DETECTION**

**Part I. Personal Data (Optional)**

Name: \_\_\_\_\_ Date: \_\_\_\_\_  
(Pangalan) (Petsa)

☐

Expert

☐

End-user

**Part II. Level of Acceptability of IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection**

Direction: Using the scale below, rate the IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection according to the criteria by ensuring that the relevant column is checked.

*(Direksyon: Gamit ang sukat sa ibaba, i-rate ang IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection ayon sa pamantayan sa pamamagitan ng pagtiyak na ang nauugnay na column ay nalagyan ng tsek.)*

Scales are as follows:

- 5 – Very Much Acceptable (*Sobrang katanggap-tanggap*)
- 4 – Much Acceptable (*Lubhang katanggap-tanggap*)
- 3 – Acceptable (*Katanggap-tanggap*)
- 2 – Slightly Acceptable (*Bahagyang katanggap-tanggap*)
- 1 – Not Acceptable (*Hindi katanggap-tanggap*)



| 1. Functional Suitability                                                                                                                                                                                                                                                                                                                                                                             | 5 | 4 | 3 | 2 | 1 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| 1.1 The developed system maintains optimal incubation conditions (e.g., temperature, humidity) throughout the incubation process.<br><i>(Ang binuong sistema ay nagpapanatili ng optimal na kondisyon ng incubation (halimbawa, temperatura at humidity) sa buong proseso ng incubation.)</i>                                                                                                         |   |   |   |   |   |
| 1.2 The YOLO deep learning model consistently and accurately detects the fertility status of chicken eggs.<br><i>(Ang YOLO deep learning model ay tuloy-tuloy at tamang natutukoy ang fertility status ng mga itlog ng manok.)</i>                                                                                                                                                                    |   |   |   |   |   |
| 1.3 The developed system monitors and logs data such as temperature, humidity, and number of fertile and infertile chicken eggs, accessible through the web application.<br><i>(Ang binuong sistema ay nagmo-monitor at nagtatala ng mga datos tulad ng temperatura, humidity, at bilang ng mga fertile at infertile na itlog ng manok, na maaaring ma-access sa pamamagitan ng web application.)</i> |   |   |   |   |   |
| 1.4 The developed system provides timely notifications or alerts for detected abnormal incubation conditions or infertile eggs.<br><i>(Ang binuong sistema ay nagbibigay ng napapanahong mga notification o alerts para sa mga na-detect na abnormal na kondisyon ng incubation o infertile na itlog.)</i>                                                                                            |   |   |   |   |   |
| 1.5 The incubator automatically switches to solar power during power interruptions and seamlessly maintains functionality without disrupting operations.<br><i>(Ang incubator ay awtomatikong lumilipat sa solar power sa oras ng mga pagkaantala sa kuryente at tuloy-tuloy na napapanatili ang functionality nang hindi naaabala ang operasyon.)</i>                                                |   |   |   |   |   |
| 2. Performance Efficiency                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 2.1 The incubator powers on and initializes its components without significant delays.<br><i>(Ang incubator ay nagbubukas at initialize ng mga bahagi nito nang walang labis na pagkaantala.)</i>                                                                                                                                                                                                     |   |   |   |   |   |
| 2.2 Sensors quickly provide readings of incubation conditions, including temperature and humidity.<br><i>(Ang mga sensor ay mabilis na nagbibigay ng sukat ng mga kondisyon ng pag-iincubate, kabilang ang temperatura at humidity.)</i>                                                                                                                                                              |   |   |   |   |   |
| 2.3 The YOLO model for fertility detection processes results promptly to minimize waiting times for users.                                                                                                                                                                                                                                                                                            |   |   |   |   |   |

|                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
| (Ang YOLO model para sa fertility detection ay mabilis na nagpoproseso ng mga resulta, para kaunti ang oras ng paghihintay ng user.)                                                                                                                                                                                                                            |  |  |  |  |
| 2.4 The incubator can handle multiple processes simultaneously, such as fertility detection and incubation condition adjustments, without performance degradation.<br>(Kayang i-handle ng incubator ang maraming proseso nang sabay-sabay, tulad ng fertility detection at pag-adjust ng kondisyon ng incubation, nang hindi naapektuhan ang performance nito.) |  |  |  |  |
| 2.5 The incubator operates efficiently without generating excessive heat or noise.<br>(Ang incubator ay efficient na tumatakbo nang hindi lumilikha ng labis na init o ingay.)                                                                                                                                                                                  |  |  |  |  |
| <b>3. Usability</b>                                                                                                                                                                                                                                                                                                                                             |  |  |  |  |
| 3.1 The incubator is simple to assemble and configure, even for users with minimal technical expertise.<br>(Ang incubator ay madaling i-assemble at i-configure, kahit para sa mga user na may kaunting teknikal na kakayahan.)                                                                                                                                 |  |  |  |  |
| 3.2 The temperature and humidity sensor readings are presented in a clear and understandable manner.<br>(Ang mga sukat ng sensor para temperatura at humidity ay naka-present sa paraang malinaw at madaling maunawaan.)                                                                                                                                        |  |  |  |  |
| 3.3 Fertility status for each egg is displayed clearly with easily distinguishable markers, such as color-coded indicators or percentage probabilities, to guide the user.<br>(Ang fertility status ng bawat itlog ay malinaw na ipinapakita gamit ang markers, tulad ng mga color-coded na indicators o percentage probabilities, upang gabayan ang user.)     |  |  |  |  |
| 3.4 Users can conveniently access, monitor, and control incubation settings through physical controls or the web-based application.<br>(Maaaring ma-access, ma-monitor, at makontrol ng mga user ang settings ng incubation sa pamamagitan ng pisikal na mga kontrol o ng web-based application.)                                                               |  |  |  |  |
| 3.5 The web-based application provides real-time updates on temperature, humidity, and fertility detection results, ensuring users always have the latest information.                                                                                                                                                                                          |  |  |  |  |

|                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| (Ang web-based application ay nagbibigay ng real-time na mga update sa temperatura, humidity, at resulta ng fertility detection, tinitiyak na laging may latest na impormasyon ang mga user.)                                                                                                                                                    |  |  |  |  |  |
| <b>4 Reliability</b>                                                                                                                                                                                                                                                                                                                             |  |  |  |  |  |
| 4.1 Temperature and humidity sensors deliver accurate readings.<br>(Ang mga sensor para sa temperatura at humidity ay nagbibigay ng tamang sukat.)                                                                                                                                                                                               |  |  |  |  |  |
| 4.2 Key components of the incubator (e.g., heaters, fans) respond effectively to maintain optimal incubation conditions.<br>(Ang ilan sa pangunahing bahagi ng incubator gaya ng heater at fan ay mabilis na tumutugon upang mapanatili ang pinakamainam na kondisyon sa pag-incubate.)                                                          |  |  |  |  |  |
| 4.3 The YOLO deep learning model consistently identifies fertility status with high accuracy, minimizing false positives and false negatives across repeated trials.<br>(Ang YOLO deep learning model ay consistent na tinutukoy ang status ng fertility nang may mataas na accuracy, binabawasan ang mga maling pagtukoy sa iba't-bang trials.) |  |  |  |  |  |
| 4.4 The YOLO model processes and delivers fertility detection results quickly, minimizing delays in decision-making during incubation.<br>(Mabilis na naipo-proseso at ibinibigay ng YOLO model ang mga resulta ng fertility detection, na iniwasan ang mga pagkaantala sa decision-making sa panahon ng incubation.)                            |  |  |  |  |  |
| 4.5 The incubator operates continuously and reliably over extended periods, with no significant failures or downtime.<br>(Ang incubator ay patuloy at maaasahang gumagana sa mahabang panahon, nang walang failiures o downtime.)                                                                                                                |  |  |  |  |  |
| <b>5 Maintainability</b>                                                                                                                                                                                                                                                                                                                         |  |  |  |  |  |
| 5.1 The incubator's design facilitates easy cleaning without requiring disassembly of key components.<br>(Ang disenyo ng incubator ay nagpapadali ng paglilinis nito nang hindi kinakailangang i-disassemble ang mga pangunahing bahagi ng incubator.)                                                                                           |  |  |  |  |  |
| 5.2 Internal wiring and connections are well-organized for easy inspection and maintenance.<br>(Ang mga wiring at koneksyon ay nakaayos para sa madaling inspeksyon at maintenance.)                                                                                                                                                             |  |  |  |  |  |

|                                                                                                                                                                                                                                                                                              |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
| 5.3 Replacement of commonly used parts (e.g., fans, heaters, sensors) is user-friendly, with components easily accessible.<br><i>(Ang pagpapalit ng mga pangunahing bahagi ng incubator, gaya ng heater, fan, at mga sensor, ay madaling gawin at ang mga bahagi ay madaling ma-access.)</i> |  |  |  |  |  |
| 5.4 Components are labeled with detailed information, ensuring easy identification for repairs or replacements.<br><i>(Ang mga components ay may label na may detalyadong impormasyon, tinitiyak ang madaling pag-identify ng kailangang repair o replacement.)</i>                          |  |  |  |  |  |
| 5.5 Maintenance tasks can be carried out using standard tools, eliminating the need for specialized equipment.<br><i>(Ang maintenance ay maaaring isagawa gamit ang mga standard na tools, hindi na kailangan ng mga specialized na equipment.)</i>                                          |  |  |  |  |  |

**Comments/Suggestions:**

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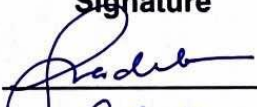
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**CERTIFICATE OF CONTENT VALIDATION**

This is to certify that the following experts have validated the researchers' questionnaire checklist prepared and submitted by Ashanty Madeline DC. Cruz, Brian Ezekiel D. Batalon, Tristan Manuel T. Santo Tomas, and Mark Jayson F. Villacura, for their undergraduate thesis entitled INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION as a partial requirement for the degree Bachelor of Science in Electronics Engineering (BSECE).

| Name                             | Signature                                                                            | Date      |
|----------------------------------|--------------------------------------------------------------------------------------|-----------|
| Engr. Jenadel DL. Antipolo       |  | 1/14/2025 |
| Engr. Rio Camille M. Pedrocillo  |  | 1/15/2025 |
| Engr. Crystaleene Jade A. Santos |  | 1/15/2025 |
| Engr. Erickson T. Marcos         |  | 1/15/2025 |



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## **LETTER TO THE RESPONDENTS**

January 13, 2025

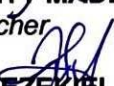
Dear Respondent,

We, the researchers, are conducting the research study entitled **INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION** in partial fulfillment of the requirements for the degree of Bachelor of Science in Electronics Engineering (BSECE). In light of this, we would like to have you as one of our respondents in this research study. We will appreciate it if you could accomplish the attached questionnaire checklist promptly and honestly.

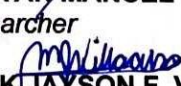
Rest assured that your response would be treated with utmost confidentiality. We are hoping for your favorable response. Thank you very much.

Respectfully yours,

  
**ASHANTY MADELINE DC. CRUZ**  
Researcher

  
**BRIAN EZEKIEL D. BATALON**  
Researcher

  
**TRISTAN MANUEL T. SANTO TOMAS**  
Researcher

  
**MARK JAYSON F. VILLACURA**  
Researcher

Noted:

  
**JENADEL DL. ANTIPOLO, ECE**  
Research Instructor & Adviser





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**REQUEST LETTER FOR PERMISSION TO CONDUCT TESTING OF DEVICE**

January 13, 2025

**Janjan Oyong**  
Hatchery Worker, Honey Red Candice Hatchery  
Brgy. Maybangcal, Morong, Rizal

Dear Sir,

Warmest Greetings!

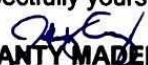
We are students of Bachelor of Science in Electronics Engineering, 4 ECE-A at the University of Rizal System – Morong Campus and are presently conducting a Capstone project entitled "IncuVision: A Computer Vision-Enhanced Chicken Egg Incubator for Fertility Detection" which is in partial fulfillment for requirements in ECE PROJ II (Design 2/ Capstone Project 2).

In connection with our study, we respectfully seek your consent to conduct device testing at Honey Red Candice Hatchery. Our developed incubator will be in your facility to assess the accuracy of our YOLO model for fertility detection of chicken eggs. The testing period will span from January to February 2025.

Rest assured, all data and information gathered during the testing will be treated with the utmost confidentiality and solely for academic purposes.

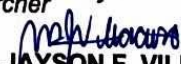
Thank you very much for considering our request. We look forward to your positive response.

Respectfully yours,

  
**ASHANTY MADELINE DC. CRUZ**  
Researcher

  
**BRIAN EZERIEL D. BATALON**  
Researcher

  
**TRISTAN MANUEL T. SANTO TOMAS**  
Researcher

  
**MARK JAYSON F. VILLACURA**  
Researcher

Noted:

  
**JENADEL DL. ANTIPOLLO, ECE**  
Research Instructor & Adviser

Approved by:  
  
**JANJAN OYONG**  
Hatchery Worker



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**APPLICATION FOR UNDERGRADUATE THESIS DEFENSE**

**Student Information**

| Last Name  | First Name       | Middle Name | Student No. |
|------------|------------------|-------------|-------------|
| CRUZ       | ASHANTY MADELINE | DELA CRUZ   | M2021-0601  |
| BATALON    | BRIAN EZEKIEL    | DELA CRUZ   | M2021-0617  |
| STO. TOMAS | TRISTAN MANUEL   | TEODORO     | M2021-0595  |
| VILLACURA  | MARK JAYSON      | FLORIN      | M2021-0638  |

**Academic Information**

|                |                                                |             |           |
|----------------|------------------------------------------------|-------------|-----------|
| Program        | BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING |             |           |
| Year & Section | 4 ECE-A                                        | SY Semester | 2024-2025 |

**Thesis Information**

|                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thesis Title                                                                                                                                                                                                                                                                                                                                                                                                                        |
| INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION                                                                                                                                                                                                                                                                                                                                          |
| Defense applied for: <input checked="" type="checkbox"/> Proposal Defense <input type="checkbox"/> Final Defense                                                                                                                                                                                                                                                                                                                    |
| Adviser's Certification<br>I hereby certify that I have read the Thesis manuscript by the students in connection with this application for proposal/final defense and classify the same as eligible for defense within the schedule/deadlines set by the university.<br><br><br>JENADEL DL. ANTIPOLO, ECE<br>Signature Over Printed Name / Date |

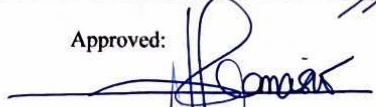
**Schedule and Panel Members**

| Date / Time of Defense              | PANEL MEMBERS  |                                                                                                                        |
|-------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------|
| October 4, 2024 /<br>11AM – 12:30PM | Chairperson:   | ENGR. RIO CAMILLE M. PEDROCILLO   |
| Assigned Room                       | Expert:        | ENGR. CRYSTALEENE JADE A. SANTOS  |
| Room 2C, SNT Building               | Critic Reader: | ENGR. ERICKSON T. MARCOS          |

Recommending Approval:

  
JENADEL DL. ANTIPOLO, ECE  
Research Instructor

Approved:

  
NORMAN C. FRANCISCO, REE  
College Research Coordinator

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**APPLICATION FOR UNDERGRADUATE THESIS DEFENSE**

**Student Information**

| Last Name  | First Name       | Middle Name | Student No. |
|------------|------------------|-------------|-------------|
| CRUZ       | ASHANTY MADELINE | DELA CRUZ   | M2021-0601  |
| BATALON    | BRIAN EZEKIEL    | DELA CRUZ   | M2021-0617  |
| STO. TOMAS | TRISTAN MANUEL   | TEODORO     | M2021-0595  |
| VILLACURA  | MARK JAYSON      | FLORIN      | M2021-0638  |

**Academic Information**

|                |                                                |              |           |
|----------------|------------------------------------------------|--------------|-----------|
| Program        | BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING |              |           |
| Year & Section | 4 ECE-A                                        | SY. Semester | 2024-2025 |

**Thesis Information**

|                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thesis Title                                                                                                                                                                                                                                                                                                                                 |
| INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION                                                                                                                                                                                                                                                   |
| Defense applied for: <input type="checkbox"/> Proposal Defense <input checked="" type="checkbox"/> Final Defense                                                                                                                                                                                                                             |
| Adviser's Certification<br>I hereby certify that I have read the Thesis manuscript by the students in connection with this application for proposal/final defense and classify the same as eligible for defense within the schedule/deadlines set by the university.<br><br>JENADEL DL. ANTIPOLLO, ECE<br>Signature Over Printed Name / Date |

**Schedule and Panel Members**

| Date/Time of Defense       | PANEL MEMBERS  |                                  |
|----------------------------|----------------|----------------------------------|
| March 26, 2025 / 1PM – 3PM | Chairperson:   | ENGR. RIO CAMILLE M. PEDROCILLO  |
| Assigned Room              | Expert:        | ENGR. CRYSTALEENE JADE A. SANTOS |
| Room 2C, SNT Building      | Critic Reader: | ENGR. ERICKSON T. MARCOS         |

Recommending Approval:

JENADEL DL. ANTIPOLLO, ECE  
Research Instructor

Approved:

NORMAN C. FRANCISCO, REE  
College Research Coordinator

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**College of Engineering - URS Morong**

**CERTIFICATE OF COMPLETION**

(Undergraduate Thesis)

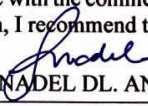
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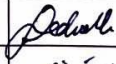
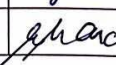
**Academic Information**

|                |                                                |               |           |
|----------------|------------------------------------------------|---------------|-----------|
| Program        | BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING |               |           |
| Year & Section | 4 ECE-A                                        | SY / Semester | 2024-2025 |

**Thesis Information**

| Thesis Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| INCUVISION: AN ENHANCED CHICKEN EGG INCUBATOR WITH COMPUTER VISION FOR FERTILITY DETECTION                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                         |
| Date/Time of Defense:<br>March 26, 2025 / 12:00 NN – 2:00 PM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Assigned Room:<br>Room 2C, SNT Building |
| <b>Adviser's Certification</b><br>I hereby certify that I have read the revised thesis manuscript presented by the students as required by the members of the Oral Examination Committee. Furthermore, I certify that the revisions are in accordance with the comments, suggestions and recommendations during the final oral defense. As such, I recommend this thesis for approval.<br><br><br>JEMADEL DL. ANTIPOLLO, ECE<br>Signature Over Printed Name / Date |                                         |

**Panel Approval**

| Panel Members | Name                             | Signature                                                                             | Date     |
|---------------|----------------------------------|---------------------------------------------------------------------------------------|----------|
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**Nurturing Tomorrow's Noblest**

URS Angono  
URS Antipolo  
URS Binangonan

Tel. 8539-9930 to 31  
Tel. 8539-9932 to 34  
Tel. 8539-9935 to 37

URS Calinta  
URS Cardona  
URS Morong

Tel. 8539-9938 to 39  
Tel. 8539-9940 to 41  
Tel. 8539-9950 to 56

URS Piliila  
URS Rodriguez  
URS Taytay

Tel. 8539-9942 to 44  
Tel. 8539-9945 to 47  
Tel. 8539-9948 to 49

## Appendix D

### Solar Set-up Computational Analysis

This computational analysis shows only the approximate values of power consumption and was used to design the IncuVision Solar Set-up.

| LOAD CONSUMPTION                           |                       |          |               |           |
|--------------------------------------------|-----------------------|----------|---------------|-----------|
| COMPONENTS                                 | WATTAGE PER COMPONENT | QUANTITY | TOTAL WATTAGE | WATT/HOUR |
| Coil Heater                                | 220W                  | 1        | 220W          | 220Wh     |
| AC Fan                                     | 30.8W                 | 2        | 61.6W         | 61.6Wh    |
| DC Fan                                     | 1.56W                 | 1        | 1.56W         | 1.56Wh    |
| LED Light Bulb                             | 7W                    | 1        | 7W            | 7Wh       |
| Egg Turner                                 | 4W                    | 1        | 4W            | 4Wh       |
| Raspberry Pi 4B                            | 15.5W                 | 1        | 15.5W         | 15.5Wh    |
| Arduino Uno R3                             | 5.5W                  | 1        | 5.5W          | 5.5Wh     |
| LED Beads                                  | 1.12W                 | 24       | 26.88W        | 26.88Wh   |
| 4 Channel Relay                            | 2W                    | 1        | 2W            | 2Wh       |
| Timer Module                               | 0.6W                  | 1        | 0.6W          | 0.6Wh     |
| Energy Meter                               | 0.5W                  | 1        | 0.5W          | 0.5Wh     |
| PSU (IDLE)                                 | 10W                   | 3        | 10W           | 10Wh      |
| Total Power/Load Consumption Per Hour      |                       |          |               | 355.14Wh  |
| System Loss                                |                       |          |               | 35%       |
| Total Power/Load Consumption + System Loss |                       |          |               | 479.439Wh |

|                                                  |     |
|--------------------------------------------------|-----|
| Battery Depth of Discharge (D.O.D) for Lead Acid | 50% |
|--------------------------------------------------|-----|

| Battery Capacity Computation               |                           |                            |                      |
|--------------------------------------------|---------------------------|----------------------------|----------------------|
| Total Power/Load Consumption + System Loss | Discharge Avider (Pt x 2) | Battery Size (Pt x 2 / Vb) | Battery Availability |
| 479.439Wh                                  | 958.878Wh                 | 79.9AH                     | 100AH                |

|                                                         |            |
|---------------------------------------------------------|------------|
| PhotoVoltaic Power Potential (PVOUT) in the Philippines | 3.6-5.2hrs |
|---------------------------------------------------------|------------|

| Solar Panel Computation              |                         |                                     |                                   |
|--------------------------------------|-------------------------|-------------------------------------|-----------------------------------|
| Battery Wattage                      | D.O.D                   | Battery Load Consumption (Pb/D.O.D) | Time of Consumption (Pb/D.O.D/Pt) |
| 1200W                                | 50%                     | 600Wh                               | 1.25hrs                           |
| PhotoVoltaic Power Potential (PVOUT) | Total Power Consumption | Solar Panel Size (Pt/PVOUT)         | Solar Panel Availability          |
| 3.6-5.2hrs                           | 479.439Wh               | 108.97W                             | 100W                              |

|                                                  |        |
|--------------------------------------------------|--------|
| <b>Solar Charge Controller Efficiency (MPPT)</b> | 90-98% |
|--------------------------------------------------|--------|

| Solar Charge Controller (MPPT) Computation |                              |                                                                                 |                                      |
|--------------------------------------------|------------------------------|---------------------------------------------------------------------------------|--------------------------------------|
| Solar Panel Wattage                        | Battery Bank Nominal Voltage | Solar Charge Controller Size (Solar Panel Wattage/Battery Bank Nominal Voltage) | Solar Charge Controller Availability |
| 100W                                       | 12V                          | 8.33A                                                                           | 20-30A                               |

| Inverter Sizing Computation |                      |                                           |                       |
|-----------------------------|----------------------|-------------------------------------------|-----------------------|
| Total Power Consumption     | Allowance Efficiency | Inverter Size (Pt + Allowance Efficiency) | Inverter Availability |
| 479.439Wh                   | 20%                  | 575.3268W                                 | 1000W                 |

The table on the next page shows the exact power consumption during the incubation process. Since many components was not always on, the power consumption was less than the approximate power consumption and the time consumption for back up power is greater than the approximate values.

| COMPONENTS POWER CONSUMPTION                              |         |          |                                |                             |
|-----------------------------------------------------------|---------|----------|--------------------------------|-----------------------------|
| COMPONENTS                                                | WATTAGE | QUANTITY | TIME USAGE                     | TOTAL WATTAGE PER HOUR      |
| Coil Heater                                               | 220W    | 1        | 5-10mins                       | 27.5Wh                      |
| AC Fan 1                                                  | 30.8W   | 1        | 5-10mins                       | 3.85Wh                      |
| AC Fan 2                                                  | 30.8W   | 1        | 1hr                            | 30.8Wh                      |
| DC Fan                                                    | 1.56W   | 1        | 1hr                            | 1.56Wh                      |
| LED Light Bulb                                            | 7W      | 1        | 1min                           | 0.1167Wh                    |
| Egg Turner                                                | 4W      | 1        | 2.5s                           | 0.0027Wh                    |
| Raspberry Pi 4B                                           | 15.5W   | 1        | 1hr                            | 15.5Wh                      |
| Arduino Uno R3                                            | 5.5W    | 1        | 1hr                            | 5.5Wh                       |
| LED Beads                                                 | 1.12W   | 24       | 30s per day every 7th/18th day | 0.0093Wh every 7th/18th day |
| 4 Channel Relay                                           | 2W      | 1        | 1hr                            | 2Wh                         |
| Timer Module                                              | 0.6W    | 1        | 1hr                            | 0.6Wh                       |
| Energy Meter                                              | 0.5W    | 1        | 1hr                            | 0.5Wh                       |
| PSU (IDLE)                                                | 10W     | 3        | 1hr                            | 10Wh                        |
| Total Power/Load Consumption Per Hour                     |         |          |                                | 97.9294Wh                   |
| Total Power/Load Consumption Per Hour During 7th/18th day |         |          |                                | 97.9387Wh                   |
| Total Power/Load Consumption Per Day                      |         |          |                                | 2.3503kW/d                  |
| Total Power/Load Consumption Per Day During 7th/18th day  |         |          |                                | 2.3505kW/d                  |
| Total Power/Load Consumption Per Incubation Process       |         |          |                                | 49.3567kW                   |

| Solar Set-up Time Consumption           |                |                 |       |                                     |
|-----------------------------------------|----------------|-----------------|-------|-------------------------------------|
| Battery Voltage                         | Battery Ampere | Battery Wattage | D.O.D | Battery Load Consumption (Pb/D.O.D) |
| 12                                      | 100A           | 1200W           | 50%   | 600Wh                               |
| Exact Time of Consumption (Pb/D.O.D/Pt) |                |                 |       |                                     |
| Regular Incubation Days                 |                |                 |       | 6.1268hrs                           |
| During 7th & 18th Day                   |                |                 |       | 6.1268hrs                           |



| Battery Charging Time  |       |                 |       |                                     |
|------------------------|-------|-----------------|-------|-------------------------------------|
| PSU                    | Solar | Battery Wattage | D.O.D | Battery Load Consumption (Pb/D.O.D) |
| 144W                   | 100W  | 1200W           | 50%   | 600Wh                               |
| Type of Supply         |       |                 |       | Charging Time (Supply Wattage/BLC)  |
| 12VDC/12A Power Supply |       |                 |       | 4.167hrs                            |
| 100W Solar Panel       |       |                 |       | 6hrs                                |

The table below shows the power consumption of the device when its charging. This was only applicable if the battery is at or below 50% and after power outage. The consumption when charging varies on how long the power outage last and the battery capacity being used.

| Total Power/Load Consumption Including Charging           |         |          |            |                        |
|-----------------------------------------------------------|---------|----------|------------|------------------------|
| COMPONENTS                                                | WATTAGE | QUANTITY | TIME USAGE | TOTAL WATTAGE PER HOUR |
| 12VDC/12A PSU                                             | 144     | 1        | 4.167hrs   | 600.048Wh              |
| Total Power/Load Consumption Per Hour                     |         |          |            | 697.9774Wh             |
| Total Power/Load Consumption Per Hour During 7th/18th day |         |          |            | 697.9387Wh             |
| Total Power/Load Consumption Per Day                      |         |          |            | 2.9503kW/d             |
| Total Power/Load Consumption Per Day During 7th/18th day  |         |          |            | 2.9505kW/d             |

## Appendix E

### Record Sheet

Record Sheet for Temperature and Humidity Monitoring Using Analog Thermo-Hygrometer Readings as Reference Values in IncuVision: An Enhanced Chicken Egg Incubator with Computer Vision for Fertility Detection

| Date             | Day | Time      | Temperature (°C) | Humidity (%) |
|------------------|-----|-----------|------------------|--------------|
| January 14, 2025 | 1   | 6:30 A.M. | 37.0             | 57.0         |
|                  |     | 6:30 P.M. | 37.5             | 57.5         |
| January 15, 2025 | 2   | 6:30 A.M. | 36.5             | 61.0         |
|                  |     | 6:30 P.M. | 37.0             | 60.0         |
| January 16, 2025 | 3   | 6:30 A.M. | 37.5             | 62.0         |
|                  |     | 6:30 P.M. | 37.5             | 58.5         |
| January 17, 2025 | 4   | 6:30 A.M. | 36.5             | 60.0         |
|                  |     | 6:30 P.M. | 37.5             | 59.0         |
| January 18, 2025 | 5   | 6:30 A.M. | 37.0             | 62.0         |
|                  |     | 6:30 P.M. | 37.0             | 64.0         |
| January 19, 2025 | 6   | 6:30 A.M. | 37.5             | 61.0         |
|                  |     | 6:30 P.M. | 37.5             | 61.5         |
| January 20, 2025 | 7   | 6:30 A.M. | 37.0             | 63.0         |
|                  |     | 6:30 P.M. | 37.0             | 59.0         |
| January 21, 2025 | 8   | 6:30 A.M. | 37.5             | 61.0         |
|                  |     | 6:30 P.M. | 37.0             | 60.0         |
| January 22, 2025 | 9   | 6:30 A.M. | 36.5             | 58.5         |
|                  |     | 6:30 P.M. | 37.5             | 64.0         |
| January 23, 2025 | 10  | 6:30 A.M. | 37.0             | 59.5         |
|                  |     | 6:30 P.M. | 37.5             | 62.0         |
| January 24, 2025 | 11  | 6:30 A.M. | 37.5             | 63.5         |
|                  |     | 6:30 P.M. | 37.0             | 63.0         |
| January 25, 2025 | 12  | 6:30 A.M. | 37.5             | 59.0         |
|                  |     | 6:30 P.M. | 37.0             | 61.0         |
| January 26, 2025 | 13  | 6:30 A.M. | 37.5             | 62.0         |
|                  |     | 6:30 P.M. | 37.5             | 58.0         |
| January 27, 2025 | 14  | 6:30 A.M. | 37.0             | 60.0         |
|                  |     | 6:30 P.M. | 38.0             | 57.0         |
| January 28, 2025 | 15  | 6:30 A.M. | 37.0             | 60.0         |
|                  |     | 6:30 P.M. | 37.0             | 62.0         |
| January 29, 2025 | 16  | 6:30 A.M. | 37.5             | 60.0         |
|                  |     | 6:30 P.M. | 37.0             | 63.0         |
| January 30, 2025 | 17  | 6:30 A.M. | 37.0             | 63.0         |
|                  |     | 6:30 P.M. | 37.0             | 59.5         |
| January 31, 2025 | 18  | 6:30 A.M. | 36.5             | 62.0         |
|                  |     | 6:30 P.M. | 37.0             | 61.0         |
| February 1, 2025 | 19  | 6:30 A.M. | 37.0             | 73.0         |
|                  |     | 6:30 P.M. | 36.5             | 71.5         |
| February 2, 2025 | 20  | 6:30 A.M. | 36.5             | 72.0         |
|                  |     | 6:30 P.M. | 37.0             | 68.0         |
| February 3, 2025 | 21  | 6:30 A.M. | 36.5             | 72.5         |
|                  |     | 6:30 P.M. | 37.0             | 71.0         |

**Record Sheet for Temperature and Humidity Monitoring Record Using DHT22  
Sensor Readings as Actual Measured Values in IncuVision: An Enhanced  
Chicken Egg Incubator with Computer Vision for Fertility Detection**

| Date             | Day | Time      | Temperature (°C) | Humidity (%) |
|------------------|-----|-----------|------------------|--------------|
| January 14, 2025 | 1   | 6:30 A.M. | 37.3             | 58.7         |
|                  |     | 6:30 P.M. | 37.4             | 59.8         |
| January 15, 2025 | 2   | 6:30 A.M. | 36.9             | 63.0         |
|                  |     | 6:30 P.M. | 36.8             | 62.7         |
| January 16, 2025 | 3   | 6:30 A.M. | 37.8             | 63.1         |
|                  |     | 6:30 P.M. | 37.3             | 59.0         |
| January 17, 2025 | 4   | 6:30 A.M. | 37.0             | 62.4         |
|                  |     | 6:30 P.M. | 37.6             | 61.1         |
| January 18, 2025 | 5   | 6:30 A.M. | 37.4             | 63.2         |
|                  |     | 6:30 P.M. | 36.9             | 61.7         |
| January 19, 2025 | 6   | 6:30 A.M. | 37.7             | 59.9         |
|                  |     | 6:30 P.M. | 37.6             | 58.6         |
| January 20, 2025 | 7   | 6:30 A.M. | 37.4             | 64.4         |
|                  |     | 6:30 P.M. | 37.2             | 59.8         |
| January 21, 2025 | 8   | 6:30 A.M. | 37.8             | 62.5         |
|                  |     | 6:30 P.M. | 37.3             | 62.6         |
| January 22, 2025 | 9   | 6:30 A.M. | 37.2             | 59.2         |
|                  |     | 6:30 P.M. | 37.6             | 63.5         |
| January 23, 2025 | 10  | 6:30 A.M. | 37.4             | 58.9         |
|                  |     | 6:30 P.M. | 37.9             | 64.4         |
| January 24, 2025 | 11  | 6:30 A.M. | 37.2             | 64.3         |
|                  |     | 6:30 P.M. | 37.4             | 64.5         |
| January 25, 2025 | 12  | 6:30 A.M. | 37.6             | 60.6         |
|                  |     | 6:30 P.M. | 37.5             | 61.5         |
| January 26, 2025 | 13  | 6:30 A.M. | 37.1             | 62.8         |
|                  |     | 6:30 P.M. | 37.8             | 60.2         |
| January 27, 2025 | 14  | 6:30 A.M. | 37.4             | 58.6         |
|                  |     | 6:30 P.M. | 37.9             | 59.1         |
| January 28, 2025 | 15  | 6:30 A.M. | 37.5             | 61.4         |
|                  |     | 6:30 P.M. | 37.3             | 64.9         |
| January 29, 2025 | 16  | 6:30 A.M. | 37.2             | 62.1         |
|                  |     | 6:30 P.M. | 36.8             | 64.5         |
| January 30, 2025 | 17  | 6:30 A.M. | 37.8             | 63.8         |
|                  |     | 6:30 P.M. | 36.9             | 62.3         |
| January 31, 2025 | 18  | 6:30 A.M. | 37.1             | 63.6         |
|                  |     | 6:30 P.M. | 36.7             | 62.5         |
| February 1, 2025 | 19  | 6:30 A.M. | 36.4             | 74.4         |
|                  |     | 6:30 P.M. | 36.8             | 73.3         |
| February 2, 2025 | 20  | 6:30 A.M. | 36.8             | 74.5         |
|                  |     | 6:30 P.M. | 37.1             | 69.2         |
| February 3, 2025 | 21  | 6:30 A.M. | 36.9             | 71.7         |
|                  |     | 6:30 P.M. | 36.7             | 68.6         |

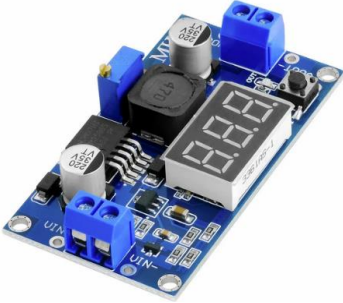
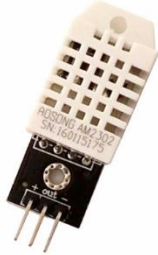


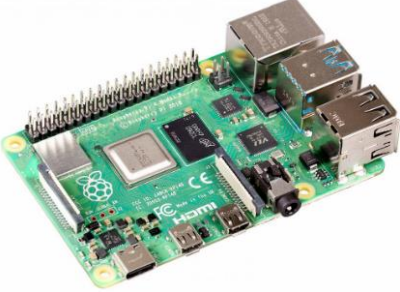
## Appendix F

### Pictorial Presentation of Different Hardware Parts and their Functions in the IncuVision

| COMPONENT NAME            | PICTURE OF COMPONENT                                                                                                                                                                               | DESCRIPTION FUNCTION                                                                                                                                 |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| AC FAN                    |  <p><a href="https://images.app.goo.gl/dCXm5TS4os57E">https://images.app.goo.gl/dCXm5TS4os57E</a><br/>MKg9</p>    | It helps regulate air circulation within the incubator, contributing to stable temperature and humidity levels essential for egg development.        |
| ADC PSU (360W)            |  <p><a href="https://images.app.goo.gl/26AQZekxpjLcaf">https://images.app.goo.gl/26AQZekxpjLcaf</a><br/>XA</p>   | Supplies stable DC power for running electronic components, ensuring the incubator and computer vision system function efficiently.                  |
| ARDUINO UNO               |  <p><a href="https://images.app.goo.gl/serZE8amabxYhS">https://images.app.goo.gl/serZE8amabxYhS</a><br/>p27</p> | A microcontroller board used for building electronic projects and automation systems.                                                                |
| AUTOMATIC TRANSFER SWITCH |  <p><a href="https://images.app.goo.gl/v6kAcUhTYCZH">https://images.app.goo.gl/v6kAcUhTYCZH</a><br/>GmwY8</p>   | Enables seamless switching between mains power and backup power sources such as solar energy, ensuring incubator reliability during power failures.. |

|                              |                                                                                                                                                                                                  |                                                                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <p>BATTERY<br/>LEAD ACID</p> |  <p><a href="https://images.app.goo.gl/95YFqZMHUHc7vnNu9">https://images.app.goo.gl/95YFqZMHUHc7vnNu9</a></p>   | <p>Stores energy from solar panels to provide backup power, maintaining optimal conditions inside the incubator during outages.</p>       |
| <p>BUTTONS</p>               |  <p><a href="https://images.app.goo.gl/J2QciKTRRMT7Z8X1A">https://images.app.goo.gl/J2QciKTRRMT7Z8X1A</a></p>   | <p>Used to manually control incubator settings, such as temperature adjustments and fertility detection system activation.</p>            |
| <p>BUZZER</p>                |  <p><a href="https://images.app.goo.gl/6LAdkpBgsdVTemts9">https://images.app.goo.gl/6LAdkpBgsdVTemts9</a></p> | <p>Provides notification alerts for critical incubator conditions, such as temperature deviations, low humidity, or hatch-ready eggs.</p> |
| <p>COIL HEATER</p>           |  <p><a href="https://images.app.goo.gl/DyGV5oh69MqR9iwB7">https://images.app.goo.gl/DyGV5oh69MqR9iwB7</a></p> | <p>Maintains the temperature inside the incubator by converting electrical energy into heat, supporting proper egg development.</p>       |

|                      |                                                                                                                                                                                                  |                                                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| DC-DC BUCK CONVERTER |  <p><a href="https://images.app.goo.gl/8tYHDEnRJS5CuU6j8">https://images.app.goo.gl/8tYHDEnRJS5CuU6j8</a></p>   | Regulates power supply to low-voltage components like sensors, ensuring stable operation of the web-based monitoring system.              |
| DHT22                |  <p><a href="https://images.app.goo.gl/J2QciKTRRMT7Z8X1A">https://images.app.goo.gl/J2QciKTRRMT7Z8X1A</a></p>   | A temperature and humidity sensor critical for maintaining the ideal incubation environment and ensuring reliable performance.            |
| FUSE                 |  <p><a href="https://images.app.goo.gl/4aePLr9egBu6PeGK9">https://images.app.goo.gl/4aePLr9egBu6PeGK9</a></p> | Protects the incubator's electrical system from overcurrent or short circuits, ensuring system reliability and maintainability.           |
| INVERTER             |  <p><a href="https://images.app.goo.gl/DyGV5oh69MqR9iwB7">https://images.app.goo.gl/DyGV5oh69MqR9iwB7</a></p> | Converts DC power from batteries or solar panels into AC power, ensuring the incubator and fertility detection system remain operational. |

|                        |                                                                                                                                                                                                                                                                                        |                                                                                                                                                                     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LCD I2C 20X4           |  <p><a href="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQ_gugk9aRjYpxWz39vTaxsz20mfPgFomB">https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQ_gugk9aRjYpxWz39vTaxsz20mfPgFomB</a></p> | Displays real-time temperature, humidity, and fertility detection results.                                                                                          |
| RASPBERRY PI 4         |  <p><a href="https://images.app.goo.gl/sGnGsiT5HHAQaSfm8">https://images.app.goo.gl/sGnGsiT5HHAQaSfm8</a></p>                                                                                        | The central computing unit that processes data from the camera, sensors, and fertility detection model (YOLOv8), enabling computer vision-based egg classification. |
| ROCKER/TOGGLE SWITCHES |  <p><a href="https://images.app.goo.gl/BVz566jHhjWLM9ZV7">https://images.app.goo.gl/BVz566jHhjWLM9ZV7</a></p>                                                                                        | Used to turn on/off incubator functions, such as candling, lightbulb, manual tilting, AC charging or solar backup power.                                            |
| RASPBERRY PI CAMERA    |  <p><a href="https://images.app.goo.gl/UUtZksAoguk8YaKv7">https://images.app.goo.gl/UUtZksAoguk8YaKv7</a></p>                                                                                       | Captures real-time images of the eggs for computer vision analysis, supporting candling and fertility detection.                                                    |

|                                                    |                                                                                                                                                                                                  |                                                                                                                                         |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <p>SOLAR<br/>CHARGER<br/>CONTROLLER<br/>(MPPT)</p> |  <p><a href="https://images.app.goo.gl/zvnAPQyWgjV9JbtQA">https://images.app.goo.gl/zvnAPQyWgjV9JbtQA</a></p>   | <p>Regulates charging and discharging of the backup power system, ensuring continuous operation of the incubator monitoring system.</p> |
| <p>SOLAR PANEL</p>                                 |  <p><a href="https://images.app.goo.gl/2rFnGBu84nAwo4ay8">https://images.app.goo.gl/2rFnGBu84nAwo4ay8</a></p>   | <p>Converts sunlight into electricity, providing a solar power backup source for the incubator to ensure uninterrupted operation.</p>   |
| <p>SWITCH TIMER<br/>RELAY</p>                      |  <p><a href="https://images.app.goo.gl/G5YYt2NQ9XytCkP19">https://images.app.goo.gl/G5YYt2NQ9XytCkP19</a></p> | <p>Automates power control for various incubator functions, such as egg incubator tilting.</p>                                          |
| <p>POWER METER</p>                                 |  <p><a href="https://images.app.goo.gl/YNBAvFcXP5js4ZDA6">https://images.app.goo.gl/YNBAvFcXP5js4ZDA6</a></p> | <p>Measures the power consumption of the incubator system, helping monitor and optimize energy efficiency for reliability.</p>          |

|                            |                                                                                                                                                                                                |                                                                                                                                             |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <p>LED BEADS</p>           |  <p><a href="https://images.app.goo.gl/bu4jzQNYUSPQjH1C9">https://images.app.goo.gl/bu4jzQNYUSPQjH1C9</a></p> | <p>Used in the candling process to shine bright light through eggs, allowing fertility detection via computer vision analysis.</p>          |
| <p>4-WAY CHANNEL RELAY</p> |  <p><a href="https://images.app.goo.gl/iHc8euotwdpr8ih69">https://images.app.goo.gl/iHc8euotwdpr8ih69</a></p> | <p>Controls multiple incubator functions, such as switching heaters, fans, or lights, supporting functional suitability and automation.</p> |

## Appendix G

### Bill of Materials

| QUANTITY | UNIT    | DESCRIPTION                                 | PRICE (PHP) | TOTAL PRICE (PHP) |
|----------|---------|---------------------------------------------|-------------|-------------------|
| 1        |         | 3W Led Beads                                | 79          | 79                |
| 1        | Piece   | 12V Switch Timer Relay Module               | 174         | 174               |
| 2        | Meter   | 40 Pin Dupont Jumper Wire Cables (30cm F-F) | 89          | 178               |
| 1        | Piece   | AC Cord                                     | 60          | 60                |
| 1        | Piece   | Acrylic                                     | 266         | 266               |
| 1        | Piece   | ADC PSU (360W)                              | 655         | 655               |
| 1        | Piece   | Analog Thermohygrometer                     | 57          | 57                |
| 1        | Piece   | Arduino Uno                                 | 549         | 549               |
| 1        | Piece   | Automatic Transfer Switch                   | 632         | 632               |
| 1        | Piece   | Battery                                     | 4488        | 4488              |
| 2        | Piece   | Bracket Connector                           | 38          | 76                |
| 1        | Piece   | Bracket Side Egg Holder                     | 40          | 40                |
| 4        | Piece   | Buttons                                     | 60          | 240               |
| 1        | Piece   | Buzzer                                      | 68          | 68                |
| 1        | Meter   | Chicken Wire (1m)                           | 55          | 55                |
| 1        | Piece   | Coil Heater                                 | 207         | 207               |
| 1        | Piece   | DC-DC Buck Converter                        | 75          | 75                |
| 1        | Piece   | DHT22                                       | 162         | 162               |
| 4        | Piece   | Door Catcher                                | 9           | 36                |
| 1        | Piece   | Egg Tray (27x20x2cm)                        | 455         | 455               |
| 2        | Piece   | Fan                                         | 145         | 290               |
| 1        | Piece   | Fuse                                        | 40          | 40                |
| 1        | Piece   | Heat Shrink                                 | 108         | 108               |
| 3        | Meter   | Insulation Foam Sheets                      | 75          | 225               |
| 1        | Piece   | Inverter                                    | 1236        | 1236              |
| 1        | Piece   | LCD Display                                 | 199         | 199               |
| 30       | Piece   | LED Beads                                   | 10          | 300               |
| 1        | Piece   | Light Bulb                                  | 71          | 71                |
| 2        | Piece   | Light Socket                                | 17          | 34                |
| 2        | Piece   | Lock                                        | 25          | 50                |
| 1        | Piece   | Raspberry Pi                                | 6989        | 6989              |
| 1        | Piece   | Ref Switch                                  | 199         | 199               |
| 4        | Piece   | Rocker/Toggle Switches                      | 30          | 120               |
| 1        | Piece   | Rotatable Mount For Solar                   | 440         | 440               |
| 1        | Piece   | RPI Camera                                  | 2552        | 2552              |
| 1        | Piece   | Silicon Sealant                             | 99          | 99                |
| 1        | Piece   | Solar Charge Controller                     | 436         | 436               |
| 1        | Piece   | Solar Panel                                 | 1669        | 1669              |
| 1        | Piece   | Support Hinge                               | 159         | 159               |
| 1        | Piece   | Terminal Block                              | 36          | 36                |
| 1        | Piece   | Tripod Stand for Solar Panel                | 450         | 450               |
| 1        | Piece   | Universal Outlet With Utility Box           | 268         | 268               |
| 1        | Piece   | Ventilation Fan                             | 100         | 100               |
| 1        | Piece   | Water Tray                                  | 99          | 99                |
| 1        | Piece   | Wattmeter                                   | 299         | 299               |
| 4        | Piece   | Wheels                                      | 22          | 88                |
| 1        | Meter   | Wire 14AWG Green (20m)                      | 230         | 230               |
| 1        | Meter   | Wire 14AWG Red (20m)                        | 230         | 230               |
| 2        | Meter   | Wire 14AWG White (5m)                       | 140         | 280               |
| 1        | Meter   | Wire 22AWG 4-Pin (10m)                      | 150         | 150               |
| 1        | Meter   | Wire Connectors                             | 147         | 147               |
| 1        | Piece   | Wire Molding PVC                            | 46          | 46                |
| 1        | Piece   | 4-Channel Relay                             | 42          | 42                |
|          |         | OTHER MATERIALS                             |             |                   |
| 1        | Piece   | Screws 300pcs                               | 65          | 65                |
| 1        | Piece   | Solder Wire 0.8mm                           | 80          | 80                |
| 3        | Piece   | Spray Paint Forest Green                    | 210         | 630               |
| 1        | Piece   | Steel Wool 20g                              | 8           | 8                 |
|          |         | FABRICATE                                   |             |                   |
| 1        | Cabinet | Custom/Owned Egg                            | 9000        | 9000              |
|          |         | <b>TOTAL:</b>                               |             | <b>36016</b>      |

## Appendix H

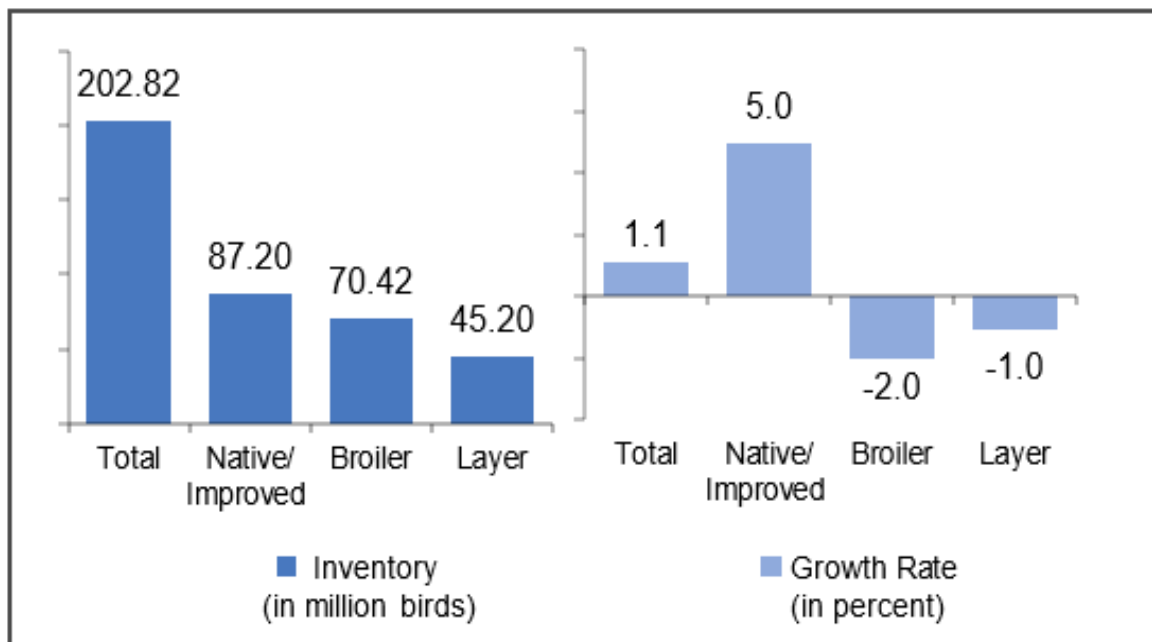
### Philippine National Standard Performance Criteria of the Egg Incubator

| Common name                  | Setting period (1st-18th day) |                 | Hatching period (19th-21st day) |                 |
|------------------------------|-------------------------------|-----------------|---------------------------------|-----------------|
|                              | Relative humidity, %          | Temperature, °C | Relative humidity, %            | Temperature, °C |
| Chicken                      | 58                            | 37.5            | 66-75                           | 36.9            |
| Duck                         | 58-62                         | 37.5            | 66-75                           | 36.9            |
| Muscovy duck                 | 58-62                         | 37.5            | 66-75                           | 36.9            |
| Quail (Bobwhite)             | 54-58                         | 37.5            | 66-74                           | 37.2            |
| Quail (Japanese)             | 58-62                         | 37.5            | 66-74                           | 37.2            |
| Turkey                       | 54-58                         | 37.5            | 66-75                           | 36.9            |
| Geese (Domestic and various) | 62                            | 37.5            | 66-75                           | 36.9            |
| Ostrich                      | 32-40                         | 36.4            | 69                              | 36.4            |



**Appendix I**

Inventory and Annual Growth Rate of Chicken by Type  
Philippines: as of 30 September 2023



## Appendix J

### Calculation of the Accuracy of Temperature and Humidity Sensor

```

allowed error - temp.py - D:/Desktop/Downloads/allowed error - temp.py (3.9.13)
File Edit Format Run Options Window Help

import numpy as np

# Define the reference temperature range
reference_temps = np.arange(36.5, 38.1, 0.5) # From 36.5 to 38.0 in steps of 0.5
deviation = 0.5 # Error based on DHT22 datasheet

# Function to calculate percentage error and accuracy
# Ensures that accuracy is evaluated within the specified ±0.5°C range
def calculate_error(reference, deviation):
    measured_min = reference - deviation # Lower bound of measured temperature
    measured_max = reference + deviation # Upper bound of measured temperature
    min_error = 0.0 # Minimum error is 0% when measured = reference
    max_error = abs(deviation / reference) * 100 # Maximum percentage error based on ±0.5°C deviation
    min_accuracy = 100 - min_error # Best-case accuracy (100%) when measured = reference
    max_accuracy = 100 - max_error # Worst-case accuracy when deviation is at maximum limit

    return reference, measured_min, measured_max, min_error, max_error, min_accuracy, max_accuracy

# Calculate errors and accuracies for all reference temperatures
results = [calculate_error(temp, deviation) for temp in reference_temps]

# Print results with clarification that accuracy aligns with DHT22 ±0.5°C specification
print("Reference Temp (°C) | Measured Min (°C) | Measured Max (°C) | Min % Error | Max % Error | Min Accuracy (%) | Max Accuracy (%)")
print("(All values computed based on the ±0.5°C accuracy range from DHT22 datasheet)")
for ref, min_temp, max_temp, min_err, max_err, min_acc, max_acc in results:
    print(f"{ref:.1f} | {min_temp:.1f} | {max_temp:.1f} | {min_err:.2f}% | {max_err:.2f}% | {min_acc:.2f}% | {max_acc:.2f}%")

# Find and print the highest and lowest possible error percentages
max_error = max(result[4] for result in results)
min_error = min(result[3] for result in results)
print(f"Highest Possible Error Percentage: {max_error:.2f}% (Based on ±0.5°C range)")
print(f"Lowest Possible Error Percentage: {min_error:.2f}% (Occurs when measured temperature = reference temperature)")

IDLE Shell 3.9.13
File Edit Shell Debug Options Window Help
Python 3.9.13 (tags/v3.9.13:6de20a5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: D:/Desktop/Downloads/allowed error - temp.py =====
Reference Temp (°C) | Measured Min (°C) | Measured Max (°C) | Min % Error | Max % Error | Min Accuracy (%) | Max Accuracy (%)
(All values computed based on the ±0.5°C accuracy range from DHT22 datasheet)
36.5 | 36.0 | 37.0 | 0.00% | 1.37% | 100.00% | 98.63%
37.0 | 36.5 | 37.5 | 0.00% | 1.35% | 100.00% | 98.65%
37.5 | 37.0 | 38.0 | 0.00% | 1.33% | 100.00% | 98.67%
38.0 | 37.5 | 38.5 | 0.00% | 1.32% | 100.00% | 98.68%

Highest Possible Error Percentage: 1.37% (Based on ±0.5°C range)
Lowest Possible Error Percentage: 0.00% (Occurs when measured temperature = reference temperature)
>>>

```

## Appendix K

### Calculation of Percent Error of Fertility Detection

```
percentage error - fertility detection.py - D:/Desktop/Downloads/percentage error - fertility detection.py (3.9.13)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# Ensure full column visibility
pd.set_option("display.max_columns", None)
pd.set_option("display.width", 1000)

# Data
data = {
    'Day': [1, 2, 3, 4, 5, 6, 7],
    'Fertile_Manual': [0, 0, 0, 0, 9, 17, 20],
    'Fertile_Auto': [0, 0, 0, 0, 13, 15, 22],
    'Infertile_Manual': [24, 24, 24, 24, 15, 7, 4],
    'Infertile_Auto': [24, 24, 24, 24, 11, 9, 2]
}

# Create a DataFrame
df = pd.DataFrame(data)

# Calculate fertile and infertile percentage errors (rounded to 2 decimal places)
df['Fertile_Error'] = np.where(df['Fertile_Manual'] == 0, 0,
                              round(abs(df['Fertile_Auto'] - df['Fertile_Manual']) / df['Fertile_Manual'] * 100, 2))

df['Infertile_Error'] = np.where(df['Infertile_Manual'] == 0, 0,
                                 round(abs(df['Infertile_Auto'] - df['Infertile_Manual']) / df['Infertile_Manual'] * 100, 2))

# Calculate overall errors (sum of errors divided by 7, rounded to 2 decimal places)
overall_fertile_error = round(df['Fertile_Error'].sum() / 7, 2)
overall_infertile_error = round(df['Infertile_Error'].sum() / 7, 2)

# Display the table
print(df[['Day', 'Fertile_Manual', 'Fertile_Auto', 'Fertile_Error', 'Infertile_Manual', 'Infertile_Auto', 'Infertile_Error']])
print(f"\nOverall Fertile Percentage Error: {overall_fertile_error}%")
print(f"Overall Infertile Percentage Error: {overall_infertile_error}%")
```

```
IDLE Shell 3.9.13
File Edit Shell Debug Options Window Help

Python 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: D:/Desktop/Downloads/percentage error - fertility detection.py =====
  Day  Fertile_Manual  Fertile_Auto  Fertile_Error  Infertile_Manual  Infertile_Auto  Infertile_Error
0    1               0             0           0.00                24             24           0.00
1    2               0             0           0.00                24             24           0.00
2    3               0             0           0.00                24             24           0.00
3    4               0             0           0.00                24             24           0.00
4    5                9            13          44.44                15             11          26.67
5    6               17            15          11.76                 7              9          28.57
6    7               20            22          10.00                 4              2          50.00

Overall Fertile Percentage Error: 9.46%
Overall Infertile Percentage Error: 15.03%
>>>
```

Ln: 8 Col: 68

## Appendix L

### Calculation of Weighted Mean for the Level of Acceptability of IncuVision

```

weighted mean - level of acceptability.py - D:/Desktop/Downloads/weighted mean - level of acceptability.py (3.9.13)
File Edit Format Run Options Window Help

import pandas as pd

# Data: [(Item Number, Ratings: 5, 4, 3, 2, 1)]
data = [
    ("1.1", [27, 5, 0, 0, 0]),
    ("1.2", [24, 8, 0, 0, 0]),
    ("1.3", [30, 2, 0, 0, 0]),
    ("1.4", [29, 3, 0, 0, 0]),
    ("1.5", [31, 1, 0, 0, 0]),

    ("2.1", [24, 7, 1, 0, 0]),
    ("2.2", [25, 7, 0, 0, 0]),
    ("2.3", [25, 7, 0, 0, 0]),
    ("2.4", [27, 5, 0, 0, 0]),
    ("2.5", [31, 1, 0, 0, 0]),

    ("3.1", [24, 7, 1, 0, 0]),
    ("3.2", [28, 3, 1, 0, 0]),
    ("3.3", [31, 1, 0, 0, 0]),
    ("3.4", [30, 2, 0, 0, 0]),
    ("3.5", [31, 1, 0, 0, 0]),

    ("4.1", [27, 5, 0, 0, 0]),
    ("4.2", [27, 5, 0, 0, 0]),
    ("4.3", [29, 3, 0, 0, 0]),
    ("4.4", [28, 4, 0, 0, 0]),
    ("4.5", [28, 4, 0, 0, 0]),

    ("5.1", [29, 3, 0, 0, 0]),
    ("5.2", [24, 8, 0, 0, 0]),
    ("5.3", [31, 1, 0, 0, 0]),
    ("5.4", [31, 1, 0, 0, 0]),
    ("5.5", [30, 2, 0, 0, 0]),
]

# Scale interpretation
scale_interpretation = [
    (4.20, 5.00, "Very Much Acceptable"),
    (3.40, 4.19, "Much Acceptable"),
    (2.60, 3.39, "Acceptable"),
    (1.80, 2.59, "Slightly Acceptable"),
    (1.00, 1.79, "Not Acceptable"),
]

# Function to compute weighted mean
def compute_weighted_mean(ratings):
    weights = [5, 4, 3, 2, 1]
    total_responses = sum(ratings)
    weighted_sum = sum(w * r for w, r in zip(weights, ratings))
    return round(weighted_sum / total_responses, 2) if total_responses else 0

# Function to determine verbal interpretation
def get_interpretation(mean):
    for min_val, max_val, interpretation in scale_interpretation:
        if min_val <= mean <= max_val:
            return interpretation
    return "No Interpretation"

# Compute values
results = []
for item, ratings in data:
    mean = compute_weighted_mean(ratings)
    interpretation = get_interpretation(mean)
    results.append([item, mean, interpretation])

# Create DataFrame
df = pd.DataFrame(results, columns=["Item", "Weighted Mean", "Verbal Interpretation"])

# Print results
print(df.to_string(index=False))

IDLE Shell 3.9.13
File Edit Shell Debug Options Window Help
Python 3.9.13 (tags/v3.9.13:6de2ca5, May 17 2022, 16:36:42) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: D:/Desktop/Downloads/weighted mean - level of acceptability.py ====
Item Weighted Mean Verbal Interpretation
1.1 4.84 Very Much Acceptable
1.2 4.75 Very Much Acceptable
1.3 4.94 Very Much Acceptable
1.4 4.91 Very Much Acceptable
1.5 4.97 Very Much Acceptable
2.1 4.72 Very Much Acceptable
2.2 4.78 Very Much Acceptable
2.3 4.78 Very Much Acceptable
2.4 4.84 Very Much Acceptable
2.5 4.97 Very Much Acceptable
3.1 4.72 Very Much Acceptable
3.2 4.84 Very Much Acceptable
3.3 4.97 Very Much Acceptable
3.4 4.94 Very Much Acceptable
3.5 4.97 Very Much Acceptable
4.1 4.84 Very Much Acceptable
4.2 4.84 Very Much Acceptable
4.3 4.91 Very Much Acceptable
4.4 4.88 Very Much Acceptable
4.5 4.88 Very Much Acceptable
5.1 4.91 Very Much Acceptable
5.2 4.75 Very Much Acceptable
5.3 4.97 Very Much Acceptable
5.4 4.97 Very Much Acceptable
5.5 4.94 Very Much Acceptable
>>>
Ln: 31 Col: 4

```

```

1 from flask import Flask, Response, render_template, url_for, jsonify, request, redirect, url_for, make_response, send_file
2 from datetime import time as dt_time, datetime
3 from serial.serialutil import SerialException
4 from typing import Callable, Tuple, Optional
5 from flask_sqlalchemy import SQLAlchemy
6 from pi_camera2 import PiCamera2
7 from ultralytics import YOLO
8 from flask_cors import CORS
9 from gpiozero import Buzzer, Servo
10 import gpiozero.pins.pigpio as ppg
11 from typing import Tuple
12 from mailer import Mailer
13 from rpi_lcd import LCD
14 from io import BytesIO
15 import random as rd
16 import adafruit_dht
17 import numpy as np
18 import subprocess
19 import threading
20 import schedule
21 import serial
22 import socket
23 import board
24 import time
25 import json
26 import cv2
27 import os
28
29 factory = ppg.PiGPIOFactory()
30
31 mail = Mailer(email="eggfertility58@gmail.com", password="vfnz iloe drup gdaa")
32
33 print(mail)
34
35 cys =
36     lcd = LCD()
37 except:
38     lcd = None
39
40 buzzer = Buzzer(17)
41 servo = Servo(25, pin_factory=factory)
42 servo.min()
43
44 app = Flask(__name__)
45 app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///task.db'
46 app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
47 db = SQLAlchemy(app)
48 CORS(app)
49
50 dht_device = adafruit_dht.DHT22(board.D4, use_pulseio=False)
51 # dht_device = adafruit_dht.DHT22(board.D17)
52
53 tuning = PiCamera2.load_tuning_file("tun708_noir.json")
54 camera2 = PiCamera2(tuning=tuning)
55 camera2V2.configure(camera.create_preview_configuration(main={"format": "H264", "size": (1440, 1440)}))
56 camera2.set_controls({"ExposureMode": "Auto", "ExposureTime": 1000000, "ColourGain": (1.0, 2.0), "AnalogueGain": 1.5, "Saturation": 1.1})
57 camera2V2.preview_configuration.main.align()
58 camera2.start()
59
60 #camera = cv2.VideoCapture(0)
61
62 port_num = 5000
63 mode_num = 9
64 mode1 = "temp"
65 mode2 = "humidity"
66 mode3 = "temp"
67 mode4 = "humidity"
68 mode5 = "temp"
69 mode6 = "humidity"
70 mode7 = "temp"
71 mode8 = "humidity"
72 mode9 = "temp"
73 mode10 = "humidity"
74 mode11 = "temp"
75 mode12 = "humidity"
76 mode13 = "temp"
77 mode14 = "humidity"
78 mode15 = "temp"
79 mode16 = "humidity"
80 mode17 = "temp"
81 mode18 = "humidity"
82 mode19 = "temp"
83 mode20 = "humidity"
84 mode21 = "temp"
85 mode22 = "humidity"
86 mode23 = "temp"
87 mode24 = "humidity"
88 mode25 = "temp"
89 mode26 = "humidity"
90 mode27 = "temp"
91 mode28 = "humidity"
92 mode29 = "temp"
93 mode30 = "humidity"
94 mode31 = "temp"
95 mode32 = "humidity"
96 mode33 = "temp"
97 mode34 = "humidity"
98 mode35 = "temp"
99 mode36 = "humidity"
100 mode37 = "temp"
101 mode38 = "humidity"
102 mode39 = "temp"
103 mode40 = "humidity"
104 mode41 = "temp"
105 mode42 = "humidity"
106 mode43 = "temp"
107 mode44 = "humidity"
108 mode45 = "temp"
109 mode46 = "humidity"
110 mode47 = "temp"
111 mode48 = "humidity"
112 mode49 = "temp"
113 mode50 = "humidity"
114 mode51 = "temp"
115 mode52 = "humidity"
116 mode53 = "temp"
117 mode54 = "humidity"
118 mode55 = "temp"
119 mode56 = "humidity"
120 mode57 = "temp"
121 mode58 = "humidity"
122 mode59 = "temp"
123 mode60 = "humidity"
124 mode61 = "temp"
125 mode62 = "humidity"
126 mode63 = "temp"
127 mode64 = "humidity"
128 mode65 = "temp"
129 mode66 = "humidity"
130 mode67 = "temp"
131 mode68 = "humidity"
132 mode69 = "temp"
133 mode70 = "humidity"
134 mode71 = "temp"
135 mode72 = "humidity"
136 mode73 = "temp"
137 mode74 = "humidity"
138 mode75 = "temp"
139 mode76 = "humidity"
140 mode77 = "temp"
141 mode78 = "humidity"
142 mode79 = "temp"
143 mode80 = "humidity"
144 mode81 = "temp"
145 mode82 = "humidity"
146 mode83 = "temp"
147 mode84 = "humidity"
148 mode85 = "temp"
149 mode86 = "humidity"
150 mode87 = "temp"
151 mode88 = "humidity"
152 mode89 = "temp"
153 mode90 = "humidity"
154 mode91 = "temp"
155 mode92 = "humidity"
156 mode93 = "temp"
157 mode94 = "humidity"
158 mode95 = "temp"
159 mode96 = "humidity"
160 mode97 = "temp"
161 mode98 = "humidity"
162 mode99 = "temp"
163 mode100 = "humidity"
164 mode101 = "temp"
165 mode102 = "humidity"
166 mode103 = "temp"
167 mode104 = "humidity"
168 mode105 = "temp"
169 mode106 = "humidity"
170 mode107 = "temp"
171 mode108 = "humidity"
172 mode109 = "temp"
173 mode110 = "humidity"
174 mode111 = "temp"
175 mode112 = "humidity"
176 mode113 = "temp"
177 mode114 = "humidity"
178 mode115 = "temp"
179 mode116 = "humidity"
180 mode117 = "temp"
181 mode118 = "humidity"
182 mode119 = "temp"
183 mode120 = "humidity"
184 mode121 = "temp"
185 mode122 = "humidity"
186 mode123 = "temp"
187 mode124 = "humidity"
188 mode125 = "temp"
189 mode126 = "humidity"
190 mode127 = "temp"
191 mode128 = "humidity"
192 mode129 = "temp"
193 mode130 = "humidity"
194 mode131 = "temp"
195 mode132 = "humidity"
196 mode133 = "temp"
197 mode134 = "humidity"
198 mode135 = "temp"
199 mode136 = "humidity"
200 mode137 = "temp"
201 mode138 = "humidity"
202 mode139 = "temp"
203 mode140 = "humidity"
204 mode141 = "temp"
205 mode142 = "humidity"
206 mode143 = "temp"
207 mode144 = "humidity"
208 mode145 = "temp"
209 mode146 = "humidity"
210 mode147 = "temp"
211 mode148 = "humidity"
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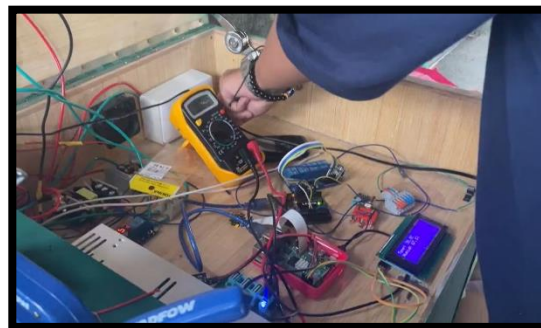
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## Appendix N

### Documentation



Chassis Construction



Wiring of the Components

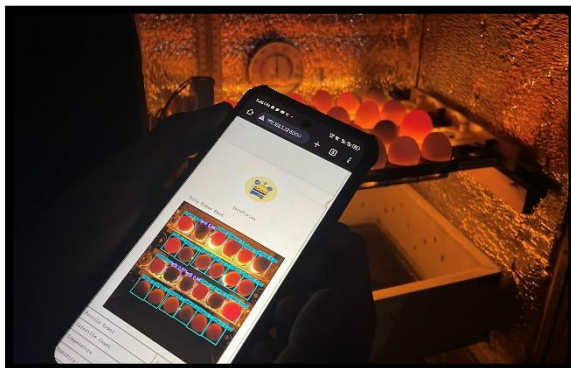


Programming and Training of the AI



Incubation Period





Fertility Detection



Testing of the Device



Conduct of Survey

# **CURRICULUM VITAE**



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---

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Teresa National High School, Science Technology Engineering and Mathematics (STEM)

#### Junior High School (2015-2019)

Teresa National High School

#### Elementary (2009 - 2015)

St. Rose of Lima Montessori School

### WORKSHOP/SEMINARS/TRAININGS

#### IECEP-URSMSC Technical Seminar (2024)

E-Connect: Establishing Transmission of Intelligence from Professionals to Future Innovators

#### Asia Open RAN Academy, DICT, USAID Webinar (2023)

Cloud or Edge: Understanding the Network Environment.

#### IECEP-RSC Webinar (2023)

Exploring Ideas and Innovating Techniques in Forming Technologically Inclined Future

#### IECEP-URSMSC Technical Seminar (2022)

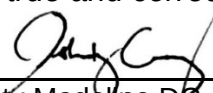
Electronics Manufacturing: Conceptualizing ECE fields through professional context.

#### IECEP-URSMSC Technical Seminar (2022)

Radio Broadcasting: Conceptualizing ECE fields through professional context.

---

*I hereby certify that the above information given are true and correct as to the best of my knowledge.*

  
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E-Connect: Establishing Transmission of Intelligence from Professionals to Future Innovators

#### Asia Open RAN Academy, DICT, USAID Webinar (2023)

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#### IECEP-URSMSC Technical Seminar (2022)

Radio Broadcasting: Conceptualizing ECE fields through professional context.

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*I hereby certify that the above information given are true and correct as to the best of my knowledge.*

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Morong National High School

#### Elementary (2009 - 2015)

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### WORKSHOP/SEMINARS/TRAININGS

#### IECEP-URSMSC Technical Workshop (2022)

Arduino Workshop 2022, Level Up: ECE for Excellence

#### IECEP-URSMSC Electronics Manufacturing (2022)

Electronics Manufacturing Technical Seminar: Conceptualizing ECE fields through professional context.

#### IECEP-URSMSC Technical Seminar (2022)

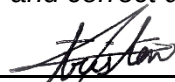
Radio Broadcasting: Conceptualizing ECE fields through professional context.

#### IECEP-URSMSC Technical Seminar (2024)

E-Connect: Establishing Transmission of Intelligence from Professionals to Future Innovators held at Science and Technology Building, University of Rizal System, Morong, Rizal.

---

*I hereby certify that the above information given are true and correct as to the best of my knowledge.*

  
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### EDUCATION

#### College (2021 - Present)

University of Rizal System - Morong Campus, BS Electronics Engineering

#### Senior High School (2019 - 2021)

Renaissance School of Science and Technology, Science Technology Engineering And Mathematics (STEM)

#### Junior High School (2015-2019)

Mahabang Parang National High School

#### Elementary (2009 - 2015)

Binangonan Elementary School

### WORKSHOP/SEMINARS/TRAININGS

#### IECEP-URSMSC Technical Workshop (2022)

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---

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---

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Student